NUMSOLVE 2.0: A CROSS-PLATFROM INTELLIGENT NUMERICAL ANALYSIS SUITE FOR EDUCATION AND RESEARCH

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Thesis submitted in partial fulfilment of the
requirement for the award of the degree of
Bachelor of Computer Science (Informatics Maritime) with Honours

FACULTY OF COMPUTER SCIENCE AND MATHEMATICS
UNIVERSITI MALAYSIA TERENGGANU
2030

THESIS CONFIRMATION AND APPROVAL

This is acknowledged and confirmed that the thesis entitled: **Numsolve 2.0: A Cross-Platform Intelligent Numerical analysis Suite for Education and Research** by **Ameer Muqrie Bin Abdul Mutalib** Matric No.: **S72433** have been checked and all the suggested corrections have been done. The thesis is submitted to the Faculty of Computer Science and Mathematics, Universiti Malaysia Terengganu in partial fulfilment of the requirements for the award of the degree of **Bachelor of Computer Science (Informatics Maritime) with Honours**.

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ACKNOWLEDGEMENTS

I would like to express my most sincere gratitude and deepest appreciation to my primary supervisor, Dr. Waheed Ali Hussein Mohammed Ghanem, for his insightful guidance, academic rigor, and the immense patience he demonstrated throughout the development of this Final Year Project. His expertise and vision were fundamental in shaping the intellectual direction of this research. I am equally indebted to my co-supervisor, Dr. Azwani Binti Alias, whose specialized knowledge in mathematics and numerical methods provided me with the necessary technical foundation and clarity required to overcome the complex computational challenges and learning curves encountered during this study. My gratitude is also extended to the academic and technical staff of the faculty for providing the essential facilities, software resources, and administrative support that made the execution of this project possible. Beyond the academic sphere, I wish to offer a special mention to my friends and fellow students, whose camaraderie, intellectual discussions, and shared experiences provided a supportive environment that made the challenging moments of this project more manageable. Most importantly, I owe my deepest debt of gratitude to my beloved family, whose unconditional love, unwavering faith in my abilities, and significant sacrifices provided the bedrock of support upon which my education is built. Their constant encouragement served as my primary source of motivation, and it is to them that I dedicate the successful completion of this academic milestone.

NUMSOLVE 2.0: A CROSS-PLATFORM INTELLIGENT NUMERICAL ANALYSIS SUITE FOR EDUCATION AND RESEARCH

ABSTRACT

NUMSOLVE 2.0: A CROSS-PLATFORM INTELLIGENT NUMERICAL ANALYSIS SUITE FOR EDUCATION AND RESEARCH addresses several critical challenges in the field of numerical analysis, including the limited accessibility of licensed software like MATLAB and the lack of intelligent guidance for students. This research aims to develop a web-based, platform-independent system that provides intelligent decision-making support and interactive visualizations to enhance the learning process for students and researchers. The methodology follows a structured Waterfall model, utilizing Java and JSP technologies for server-side logic and real-time computation, supported by a three-tier architecture and MySQL database for data management. Key outcomes include the integration of a Rule-Based Decision Support Module that recommends optimal numerical techniques based on user input, as well as real-time graphing for convergence and error analysis. The significance of this project lies in its ability to bridge the gap between theoretical mathematical concepts and practical computational application, providing a lightweight, accessible, and comprehensive alternative to commercial tools while fostering a deeper conceptual understanding through guided learning.

NUMSOLVE 2.0: SUIT ANALISIS BERANGKA PINTAR PELBAGAI PLATFORM UNTUK PENDIDIKAN DAN PENYELIDIKAN

ABSTRAK

NUMSOLVE 2.0: SUIT ANALISIS BERANGKA PINTAR PELBAGAI PLATFORM UNTUK PENDIDIKAN DAN PENYELIDIKAN menangani beberapa cabaran kritikal dalam bidang analisis berangka, termasuk kebolehcapaian terhad bagi perisian berlesen seperti MATLAB dan kekurangan panduan pintar untuk pelajar. Penyelidikan ini bertujuan untuk membangunkan sistem berasaskan web dan bebas platform yang menyediakan sokongan pembuatan keputusan pintar serta visualisasi interaktif bagi meningkatkan proses pembelajaran untuk pelajar dan penyelidik. Metodologi kajian mengikut model Waterfall yang teratur, menggunakan teknologi Java dan JSP untuk logik bahagian pelayan dan pengiraan masa nyata, disokong oleh seni bina tiga peringkat dan pangkalan data MySQL untuk pengurusan data. Hasil utama merangkumi integrasi Modul Sokongan Keputusan Berasaskan Peraturan yang mencadangkan teknik berangka optimum berdasarkan input pengguna, serta pemetaan masa nyata untuk analisis penumpuan dan ralat. Kepentingan projek ini terletak pada keupayaannya untuk merapatkan jurang antara konsep matematik teori dan aplikasi pengiraan praktikal, menyediakan alternatif yang ringan, mudah dicapai, dan menyeluruh kepada alatan komersial di samping memupuk pemahaman konsep yang lebih mendalam melalui pembelajaran berpandu.

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LIST OF ABBREVIATIONS

FTKKI	Faculty of Ocean Engineering Technology and Informatics.
GCD	Greatest Common Divisor.
LCM	Least Common Multiple.
SaaS	Software as a Service.
UMT	Universiti Malaysia Terengganu.

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CHAPTER 1

INTRODUCTION

In this chapter, we explore the framework for the NumSolve 2.0 project, which will provide the basis for the need for a numerical analysis suite that will provide intelligent interoperability across platforms. It is intended to be a comprehensive discussion of the main problems that existed in other numerical computational tools and methods, as well as a description of the specific goals, objectives and scope of the project aimed at overcoming these issues using a web-based architecture employing Java and JSP. Finally, this chapter will lay out the overall significance of the project by presenting its structural elements; this gives the reader an overview of what to expect in the next phases of technical analysis and design related to the project.

1.1 Project Background

Numerical analysis is a key part of modern science and engineering. It gives us the iterative algorithms we need to solve complicated math problems that don't have exact analytical solutions. There is a big shift in education right now toward interactive digital learning platforms that make these complicated methods easier for students to understand. But industry-standard tools like MATLAB can be hard to get into because they cost a lot to license, need a lot of system resources, and have a steep learning curve, which can make them hard to use in schools with limited resources.

This research is timely and essential as it aims to democratize access to numerical computation via a lightweight, web-based platform. The project is an improvement over NumSolve 1.0, an earlier version that used MATLAB but was limited by its dependence on the platform and the small number of methods it could use. NumSolve 2.0 is a cross-platform solution that doesn't need to be installed on a local machine. It uses a Java and JSP-based architecture, so students and researchers can do calculations from any standard web browser.

The importance of this study is that it will affect how well schools work and how much research gets done. The system connects complicated theoretical ideas with real-world use by adding an intelligent Rule-Based Decision Support Module. This module helps users choose the best numerical methods for their specific problems. Ultimately, this project adds to what we know about educational technology by giving us a complete set of tools that includes guided learning, data management, and real-time visualization all in one easy-to-use space.

1.2 Problem Statements

A modern numerical analysis platform should ideally be a flexible, easy-to-use, and smart teaching tool that not only does high-precision calculations but also walks the user through the iterative logic of each algorithm. This kind of system should let people access it from any platform without any problems, show convergence behavior in real time with graphs, and have strong data management tools that let people store and report results. But the current set of numerical tools doesn't meet these standards, as shown by the previous version, NumSolve 1.0.The earlier system was made in MATLAB, but it was very limited because it only worked on certain platforms and used a small number of methods. It only used basic root-finding and curve-fitting techniques like Bisection and Spline Interpolation. Also, many students and teachers can't use industry-standard tools like Python and MATLAB because they need a lot of resources and cost a lot to license. This is especially true in schools that don't have a lot of resources and need a lightweight, browser-based solution.

There is a big gap in research because most current computational engines work like "black boxes," giving final results but not the "intelligent guidance" that beginners need to figure out whether to use differentiation, interpolation, or specific root-finding methods on a problem. These systems usually don't have the integrated learning support that is necessary for mastering concepts, like theoretical justifications and step-by-step visualizations. Also, because there are no integrated database systems, users can't save, get back, or compare past calculations, which means they lose important data and their research isn't as efficient. Without addressing these gaps, the learning experience of a student studying numerical analysis will continue to be a confusing and difficult technical challenge, leading to more time needed to learn and a higher chance for errors in the various types of numerical analysis. Therefore, there is a need for the creation of NumSolve

2.0. This project will fill these gaps through the addition of a rule-based decision module, comprehensive classroom management tools, and report functionality in both CSV and PDF format to give the modern classroom an enhanced overall solution.

1.3 Problem Solution

NumSolve 2.0 will be utilized to address the previously stated challenges. NumSolve 2.0 is a new web-based application suite that allows numerical computations that have been previously costly and limited in manning and training, to be transitioned into an accessible and supportive learning environment. Additionally, NumSolve 2.0 will overcome the main limitation of platform dependence and expensive license fees for commercial tools such as MATLAB by being built on a web-based Java/JSP architecture. Thus NumSolve 2.0 will be available cross-platform and fully browser-based, with no need for local installations/extra hardware to access. By offering a lightweight, open access solution, NumSolve 2.0 will effectively bridge the accessibility gap for both educators and students in resource limited locations.Other than significantly increasing the computational capability of the software via many more advanced computation methods (in addition to root-finding), such as Newton-Raphson, False Position, Lagrange and Spline Interpolation, Simpson's Rule for Integration, and Solving Ordinary Differential Equations, NumSolve 2.0 will provide a comprehensive platform for both higher level education and research.

NumSolve 2.0 uses a Rule Based Decision Support Engine (RBDS) to provide smart help for beginners in order to overcome the obstacles associated with their ability to learn mathematical techniques due to a lack of intelligent guidance. The RBDS engine will give users a recommendation on how to solve problems by examining datasets or equations provided by the user and suggesting the best numerical method and technique to use; for example, suggesting interpolation methods for discrete data points, or suggesting certain root finding methods for non-linear equations. The "intelligent cure" to this problem is then enhanced by the addition of Interactive Visualization and Learning Support, which transforms the "black box" output that traditional solvers provide into a real-time graphical representation of the convergence process, along with an iterative logical flow of the steps taken to solve each problem. The inclusion of integrated mathematical formulas, as well as theoretical justification for each of the methods and techniques provided, allows for the knowledge gap between theory and practice to be successfully bridged.

At last, the solution has ended the previously encountered problems associated with fragmented learning and data loss using double integration of Data Management and Integrated Institutional Features. Many current solutions do not have any storage capabilities while NumSolve 2.0 has a fully functional database management system allowing for all CRUD (create/read/update/delete) operations enabling the creation, retrieval, and comparison of past calculations. In addition, with the capability of exporting results in either CSV (e.g., Excel) or PDF (e.g., Word) formats, there is an opportunity to produce professional reports. Additionally, to provide an optimal learning environment for students in the classroom, NumSolve 2.0 has added the Class Management Module and the Quiz/Interactive Assessment Module features to ensure that the educator can establish digital class groupings and all students can receive immediate feedback as evidence of conceptual mastery over the topics studied. By providing multiple layers of support, NumSolve 2.0 has evolved beyond simply being a calculator to become a comprehensive, advanced alternative to the many real-world technical and educational resources currently missing from numerical analysis applications.

1.4 Objectives

The objectives of this project are established to ensure the development of the NumSolve 2.0 system meets its intended goals. The objectives are arranged in sequence to reflect the systematic process from analysis to system implementation.

- To analyze the limitations of the previous NumSolve 1.0 system and existing numerical computing tools to identify specific requirements for intelligent guidance, accessibility, and expanded computational methods.
- To design a web-based, platform-independent system architecture that integrates a rule-based decision engine, a quiz-based assessment module, and a classroom management interface to improve learning engagement and usability.
- To develop the NumSolve 2.0 suite using Java and JSP technologies, incorporating real-time interactive visualizations and automated data reporting to provide a comprehensive tool for numerical analysis.

1.5 Project Scope

The project's scope defines the operational and technical limits of NumSolve 2.0, making sure that the research stays on track and can be achieved in the timeframe specified. The subsequent paragraphs of this section define what system features will be included, where the system has technical limitations, and the target audience users to eliminate the possibility of some parts of the user community having false hopes about the system's capabilities.

1.5.1 Scope Of The System

NumSolve 2.0 has been created as a web-based numerical computing package utilizing both Java and JSP technologies. The system is capable of implementing a complete set of numerical methods including the Bisection method, Secant method, Newton-Raphson method, and False Position method for finding roots; Lagrange and Spline interpolation; and numerical differentiation and integration using the Trapezoidal Rule and Simpon's Rule. In addition, the system includes hardware and software components installed on a remote computer (i.e. browser) in order to function as an ODE solver. A significant limitation for the project is that it requires access to the internet for accessing the application thus eliminating the need for other software programs (e.g., MATLAB or Python).

The Decision Support Module makes decisions based on the Rules, defined by Users; the User Inputs are analyzed to recommend a Method to use for a Problem. The Module has additional features to analyze the Method and display real-time graphical representations of both the Method's Convergence Behaviour and the Estimated Error associated with the Method. Full CRUD (Create, Read, Update, Delete) functionally for database management is incorporated into the Module to allow Users to save past calculations and to export results in either CSV or PDF format. The Module has additional functionality and resources that can be utilized to enhance the Educational Value of the Module through an Educators Class Management Module and an Interactive Quiz Module to evaluate student progress.

To help create a realistic implementation of the above development cycle, it is proposed that items not included in this project will be excluded; specifically, the scope of the project shall be limited to only one product - Web Application. There will be no native Android/iOS Mobile Application products developed since this application was developed as mobile web browser compatible and the application's functionality is configured for this purpose. In addition, this application is not designed to function off-line (thus real time, or multi-user collaboration using this application) and will not include connections with third-party environments (including, but not limited to MATLAB, Python, etc.). Although this application utilizes rule-based logic for a decision-making module, there will be no ability to utilize advanced Machine Learning (ML) and/or Artificial Intelligence (AI) algorithm(s). Finally, "Real-Time", "Multi-User", and "Cloud-based" file-sharing capabilities for this application are outside the scope of work for this application release.

1.5.2 Coverage Of The System

NumSolve 2.0 is created with a purpose of addressing both academic and research institutions around the world. Its geographic area encompasses any location where there are people with computers connected to the Internet. however, its overall focus is on higher education institutions offering Computer Science and/or Applied Mathematics degree programs. In addition, ALL interfaces and documentation created for use with this system will be developed exclusively in English due to consistency with global technical and academic requirements.

The target audience for NumSolve therefore consists of three major groups:

- **Student :** Who utilize the platform to perform computations, visualize algorithm behavior, and complete interactive assessments.
- **Educators or Lecturers :** Who use the suite as a teaching tool to demonstrate numerical techniques, manage digital classrooms, and track student progress.
- **Researchers :** Who require a reliable tool to analyze data, compare algorithm performance, and export results for academic studies or publications.

1.6 Thesis Outline

The overall structure of the research and development process used to create the NumSolve 2.0 project can be seen in the following arrangement of seven chapters, which organizes all facets of the project. Chapter one introduces the NumSolve 2.0 project by interpreting the research background, examining the previous limitations of the project, and establishing the objectives, scope and significance of the study. Chapter two provides a literature review that examines the fundamental theories of numerical methods and analyses the works of others with similar problems or projects, and the tools currently available to solve similar or related problems. Chapter three describes the research methodology that was employed to develop the project and discusses the Waterfall model, the project planning schedule, and milestones used throughout the development process.

Chapter four discusses the requirements for the entire system, and includes functional/non-functional requirements, use case diagrams, and activity diagrams developed during the requirement elicitation phase. Chapter five provides the design documents for the entire system and includes system architecture, database design, and interface design documents required to implement applications. Chapter six deals with implementation of the whole system, including the development of the hierarchical menu, the development of the core computational modules, and the development of the rule-based decision engine. Finally, chapter seven provides a summary of studies conducted to complete the project, the limitations of the project, and recommendations for future work and enhancements to the system.

CHAPTER 2

LITERATURE REVIEW

In this chapter, there is a critical assessment regarding the underlying theories and present technology that supports the creation of NumSolve 2.0. Identifying theory-based principles, i.e., iterative numerical methods and three-tier web architecture, as well as the body of works available in MATLAB, offers insight into the primary gaps in research. Consequently, synthesizing these mathematical theories with software engineering provides an academically sound foundation to support the technical approach for the project, making the proposed solution more comprehensible for the reader.

2.1 Introduction

The beginning of Chapter 2 is intended to confirm the information that has been presented in conjunction with the development of NumSolve 2.0 for the purpose of furthering the development of the project, which includes theories, technology, and previously conducted research. Chapter 1 focused on the reasons and objectives for conducting this study; whereas, Chapter 2 will critically evaluate the mathematical and engineering principles that act as the technical underpinnings of the software.

Establishing a theoretical foundation will assist in identifying the specific technological means of addressing the accessibility and educational deficiencies associated with numerical analysis tools. In essence, this section serves to establish the significance of these theories as prerequisite information prior to the detailed evaluation of user/system requirements and design. In the end, the literature review will ensure that NumSolve 2.0 contains a solid foundation of established scientific and engineering principles so as to create a connection between abstract mathematical theories and functional web-based software.

2.2 Fundamental Theory and Concept

The theoretical and architectural foundations of the numerical analysis suite must be established in order to create an intelligent and reliable numerical analysis suite. For the purposes of this study, iterative numerical algorithms will be the basis of NumSolve 2.0 and the underlying web structure will be three tiers. Within the framework of this study, the term "Concept" refers to specific variables associated with both the system's level of intelligence, such as algorithmic convergence, and its degree of educational usefulness, such as pedagogical scaffolding. Consequently, the application has been created from established evidence of science and technical standards. Each of the following sub-sections will discuss the foundational theories or concepts related to this study.

2.2.1 Theory of Numerical Iteration and Error Analysis

NumSolve 2.0 has a solid mathematical basis that utilizes the theory behind iterative numerical techniques for solving a problem. Whereas analytical mathematics uses algebra to create a closed form (an exact solution), numerical analysis accepts that for many complex problems, "successive approximations" will provide an acceptable solution. The application of bracketing methods (such as the Bisection and False Position methods) as well as non-bracketing methods (like the Newton-Raphson and Secant methods) allow for many different approaches to arrive at an approximate solution using iterative methods. One important area of study within this theory is Error Analysis; this section includes the investigation of Absolute Error, Relative Error, and Percentage Error. This study is an essential foundation for determining the accuracy of NumSolve 2.0's converging iterative logic within user-defined tolerances.

2.2.2 Three-Tier Web Architecture Theory

The project is based upon the Three Tier Client Server Architectural Development Theory, which divides the application into three separate tiers: Presentation Layer (JSP User Interface), Logic Layer (Java Servlet Mathematical Computation), and Data Layer (MySQL Persisting Data). The purpose of this research to utilize a Three Tier Client Server Architecture, which eliminates the issue of platform dependency; therefore, providing flexibility to move high level numerical computations from a desktop bound environment to a browser-based environment that increase accessibility to researchers and students.

2.2.3 Concept of Rule-Based Decision Logic

NumSolve 2.0 uses the 'intelligent' part of this system using what are called Rule Based Systems. This theory states that humans make very complex decisions, such as determining which math formula works best for a particular situation, can be found using a series of "if-then" rules. In this particular project, the rule-based system provides additional guidance to those who would typically do their own research. In the case of NumSolve, the rule based engine looks at the characteristics of your specific problems (discrete data, continuous data) and suggests an appropriate algorithm. This is key to showing that a guided set of applications is more effective for learning than a standard "black box" calculator.

2.2.4 Theoretical Framework for Formative Assessment and Classroom Management

NumSolve 2.0 combines Formative Assessment and Classroom Management and Beyond Calculation. In this research study, as a component of NumSolve 2.0, quizzes and interactive feedback of formative assessments to measure, in real-time, how well a student is learning, are used not only as a technical feature but also conceptually in assessing a student's mastery of numerical methods (Conceptual Mastery). Furthermore, the Classroom Management component of NumSolve 2.0 allows for the management of user roles to create a controlled, data-driven teaching and learning environment. This clarification of each definition supports the development of non-mathematical modules, which when completed, serve as an essential part of the complete educational suite.

2.3 Related Works

Existing tools and systems that are technically relevant to the suggested **NumSolve 2.0** system are covered in this chapter. The development of the suggested approach was motivated by the analytical, visualization, and numerical computation capabilities offered by these systems. Every linked effort is examined according to its goals, capabilities, constraints, and applicability to this project.

2.3.1 MATLAB

A popular high-level computing environment for developing algorithms, visualizing data, and doing numerical calculations is called MATLAB. It is an industry standard for numerical analysis and simulation because of its broad support for mathematical modeling and data analysis. However, MATLAB depends on high-performance technology, requires a premium subscription, and uses a lot of storage space. Additionally, students or institutions with insufficient resources cannot access it. Although MATLAB provides great computational accuracy, its exorbitant cost and complexity led to the creation of **NumSolve 2.0** as a more accessible and lightweight substitute.

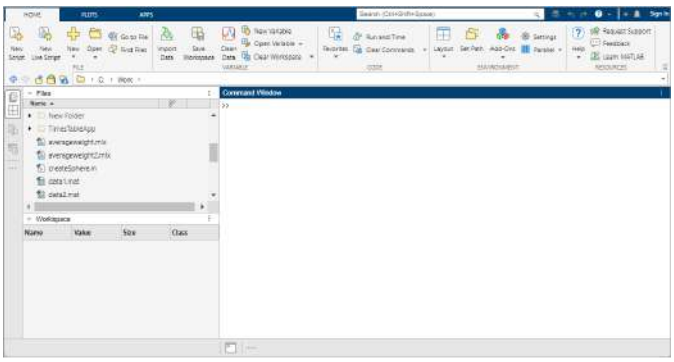


Figure 2.1: MATLAB Interface

2.3.2 Wolfram Alpha

Equations, differentiation, integration, and matrix operations are among the mathematical issues that can be resolved by the online Wolfram Alpha computational engine. It is a helpful teaching tool because it offers step-by-step instructions together with instant results. However, it does not offer user-controlled execution of numerical methods or iterative visualizations and is fully dependent on internet connectivity. Users cannot change parameters such as tolerance and iteration count. By providing an web-based system with user-controlled numerical execution with dynamic visualization and customizable input settings, **NumSolve 2.0** overcomes these drawbacks.

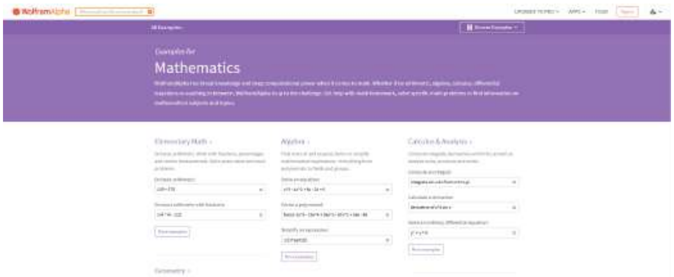


Figure 2.2: Wolfram Alpha Interface

2.3.3 GeoGebra

For instructional purposes, GeoGebra is an interactive mathematical program that blends geometry, algebra, and calculus. Although it has good graphing and visualization features, it is devoid of numerical computing tools like interpolation and root-finding techniques. GeoGebra prioritizes mathematical visualization above the use of numerical methods. **NumSolve 2.0** offers both computation and visualization capabilities in a single system by fusing the computational power of MATLAB with the interactive nature of GeoGebra.

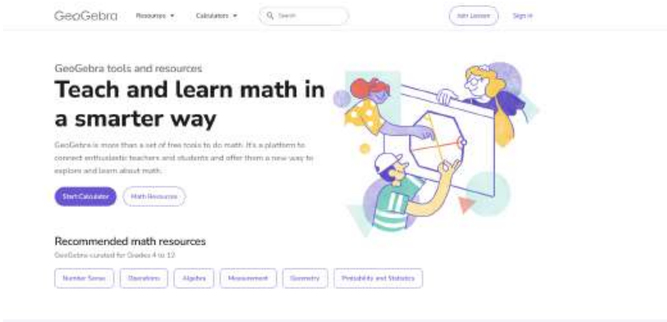


Figure 2.3: GeoGebra Interface

2.3.4 Scilab

An open-source platform for numerical computation called Scilab was created to offer features akin to those of MATLAB. It provides development environments for algorithms, graphical visualization tools, and a wealth of mathematical libraries. Numerous numerical operations, including matrix manipulation, interpolation, and root identification, are supported by Scilab. However, using it frequently necessitates programming experience, and inexperienced users may find the interface confusing. Furthermore, Scilab does not offer intelligent support to help users choose the best numerical techniques. By offering a rule-based decision module and an accessible graphical interface, **NumSolve 2.0** gets beyond these restrictions and makes problem-solving easier for both teachers and students.

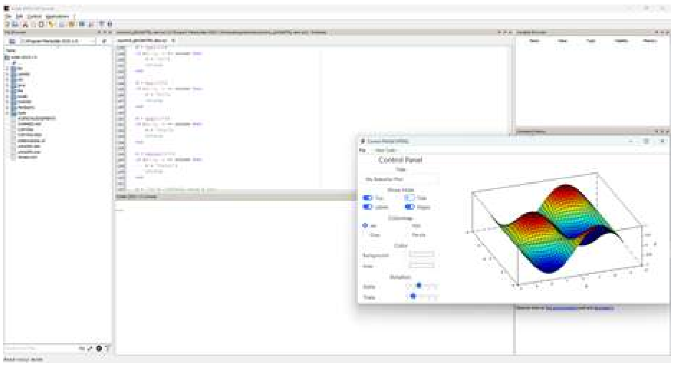


Figure 2.4: Scilab Interface

2.3.5 Symbolab

Symbolab is An online tool for solving algebraic and calculus-based issues. It is a well-liked learning and revision tool among students since it offers detailed explanations and graphical output. Symbolab, on the other hand, emphasizes analytical computation and symbolic solving over iterative numerical techniques. Additionally, it allows little parameter modification and needs an online connection to work. On the other hand, **NumSolve 2.0** is a completely offline system that emphasizes intelligent decision-making, visualization, and numerical algorithms for improved computation and learning.

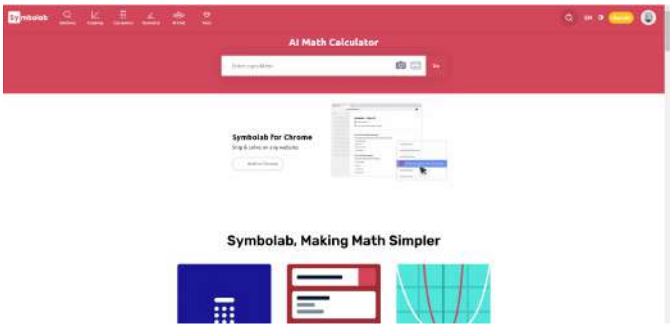


Figure 2.5: Symbolab Interface

2.3.6 Matlab 1.0

As a teaching tool for displaying fundamental numerical techniques including bisection, secant, and spline interpolation, NumSolve 1.0 (MATLAB-Based System) was created with MATLAB App Designer. Students were able to enter equations and see graphical output that showed iteration steps, error reduction, and convergence. Aside from lacking capabilities like data storage, intelligent guidance, and sophisticated techniques like Newton-Raphson, False Position, Differentiation, Integration, and ODE solvers, its reliance on MATLAB licensing restricted accessibility. NumSolve 2.0 overcomes these constraints by migrating to a web-based architecture using Java and JSP, resulting in a cross-platform, totally independent system. The project is transformed from a simple instructional tool into a full and sophisticated computational platform by adding a Rule-Based Decision Support Module, database administration, and a wider variety of numerical approaches.

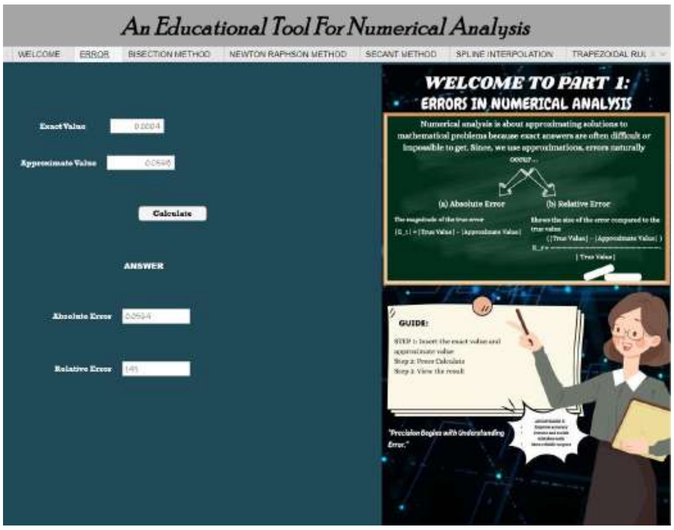


Figure 2.6: Wolfram Alpha Interface

2.3.7 Summary

Table 2.1 presents a comparison between the reviewed systems and the proposed **NumSolve 2.0**. The comparison focuses on several key criteria such as cost, platform dependency, internet requirement, learning support, visualization, and intelligent assistance.

Table 2.1: Comparison of Related Systems and NumSolve 2.0

Feature / System	MATLAB	Wolfram Alpha	GeoGebra	Scilab	SymboLab	NumSolve 1.0	NumSolve 2.0
License Cost	Paid	Free(Online)	Free	Free	Free(Online)	Free	Free
Platform Type	Desktop	Web-Based	Desktop / Web	Desktop	Web-Based	Desktop	Web-Based
Internet Required	No	Yes	Yes	No	Yes	No	Yes
Programming Knowledge Needed	Yes	No	No	Yes	No	No	No
Numerical Computation	Yes	Yes (limited)	No	Yes	No	Yes (limited)	Yes
Graphical Visualization	Yes	Yes	Yes	Yes	Yes	Yes	Yes(enhanced)
Learning Support	Partial	Yes	Yes	No	Yes	Yes	Yes(integrated)
Intelligent Guidance	No	No	No	No	No	No	Yes
Offline Accessibility	Yes	No	Yes	Yes	No	Yes	No

Each current system has unique benefits but also significant drawbacks, as indicated in Table 2.1. Although it requires programming skills and a paid license, MATLAB offers robust numerical computing and visualization capabilities. Although Wolfram Alpha and Symbolab are available online, they provide very little computational control and are totally dependent on internet availability. While Scilab facilitates calculation but requires programming knowledge and provides little learning support, GeoGebra primarily concentrates on visualization. Some of these problems were resolved by the previous NumSolve 1.0, which offered offline numerical visualization but was still reliant on MATLAB and only supported a few fundamental techniques.

The proposed system, NumSolve 2.0, is an integration of the strengths of current numerical solutions into one platform. The system is based on an internet platform and is capable of offering extensive numerical solutions, intelligent assistance, improved visualization, and learning capabilities without the aid of any commercial software. Thus, the proposed system is flexible, user-friendly, and academically appropriate.

2.4 Discussion

When considering the theories and evaluations of prior work to develop NumSolve 2.0, there is an evident gap amongst numerical analysis tools today. Most numerical analysis tools currently used in industry (MATLAB and Scilab) have high computational capabilities, but they often present a "high barrier to entry," with expensive licenses and complex programming requirements. In contrast, some web-based engines (like Wolfram Alpha and Symbolab) are simple to use, but provide only a "black box" solution at the end of the calculation, giving the user no control over any iterations (such as tolerance, step-size).

On the basis of this information, the design of a modern suite of educational tools should incorporate three distinct criteria of accessibility, and provide intelligent support to facilitate successful completion of calculations. NumSolve 2.0 addresses these three criteria through a Three-Tier Web Architecture that eliminates the platform dependency found in both NumSolve 1.0 and MATLAB by allowing users to conduct high-level calculations through a standard web browser.

The use of Rule-Based Decision Logic and Theoretical Framework for Formative Assessment and Classroom Management is the primary distinction between this project and all other current projects. Unlike other tools (such as GeoGebra) which have only a visualization focus, NumSolve 2.0 closes the gap between theory and practice by providing specific suggestions on how to achieve the program's desired outcomes based on user data. This combined approach of mathematics rigor, flexibility of the software and features that are available from the institution (such as Class Management) creates a powerful framework that meets the educational deficiencies that were identified in past research.

2.5 Summary

In conclusion, Chapter 2 has presented an extensive exploration of core theories and concepts as well as current existing systems that are directly related to creating NumSolve 2.0. This has included definitions of terminology in this research (i.e., terms such as iterative convergence, error analysis, and structural benefits from having a Java-based web architecture).

The comparative analysis of previous works established that even though all previous tools showed some capability for computations, none offered fully integrated "intelligence" and classroom-based features that form a complete environment for learning. The development of both rule-based logic and formative assessment creates a theoretical rationale for unique characteristics of the modules within the overall system. As a result, this project will evolve from a basic calculator into a guided educational tool. Together, these findings provide a strong basis for bridging the gap between theoretical mathematics and the practical application of computer software engineering. The theoretical constructs developed in this chapter will be used in Chapter 3: Methodology, which will describe the systematic Waterfall-theory format used for developing and creating these concepts.

CHAPTER 3

METHODOLOGY

The NumSolve 2.0 system's development process is described in this chapter. It describes the hardware and software requirements necessary to guarantee the system operates well, as well as the software development model and system architecture. From requirement analysis to system construction and testing, the methodology offers a well-organized framework that directs the undertaking.

3.1 Introduction

To build the new version of NumSolve 2.0, an organized and methodical approach to the design and creation of the program must be followed in order for the demanding mathematical requirements to be effectively transformed into a reliable Internet-based application. This chapter describes both the research method and technical framework used to develop and implement the project from concept through deployment. The methodology creates a blueprint for the entire development process, ensuring that the final product is computationally correct, educationally effective, and accessible by the intended user group.

The establishment of a clearly defined Software Development Life Cycle (SDLC) and clearly defined system architecture reduces the amount of risk that exists with the potential for project failure due to technology-related issues and ensures that all goals of the project will be completed within the expected timing. This chapter will describe how the Waterfall model was chosen to be the development methodology, how the development will be implemented using a Three-Tier architecture for multi-platform support, and what is needed in terms of both hardware and software tools to build the smart components of NumSolve 2.0. This methodology ultimately provides the basis for developing a suite of programs that will bridge the gap between the theoretical side of numerical analysis and the practically user-oriented software side.

3.2 Project Methodology

The methodology for the NumSolve 2.0 Project is the basis for the operation of the project (the systematic process) in turning a conceptual model (an idea) into a working software product. This section describes the various steps of selecting and following a structured developmental life cycle, as well as selecting the technical architecture that supports the core functionality of the NumSolve System. By using these established engineering standards, the project can deliver a suite of programs that are dependable, can be maintained, and have been designed specifically for meeting the unique and demanding requirements of numerical calculations, as well as the delivery of educational materials.

3.2.1 Software Model Using Waterfall

The **Waterfall Model** is the methodology used in Software Development Life Cycle (SDLC) for **NumSolve 2.0** software development. This model has been chosen for its structured approach in completing each stage before moving forward with the next one in line. This model is suitable for this project since it is certain that system requirements have been well defined and will remain constant during development.

The **Requirements Phase** consists of the following activities: user and system requirements identification and analysis, and their documentation by means of conducting research and consulting with various stakeholders. The user and system requirements are documented in a document called Software Requirements Specification (SRS).

The **Design Phase** primarily deals with the overall system architecture and detailed system design specifications. This includes database schema design, user interface structure, and computational flow for numerical methods like Bisection method, Newton-Raphson method, and numerical integration. Both High Level Design (HLD) and Low Level Design (LLD) are done so that the system and its components are well understood.

The **Programming Phase** includes the implementation of the system through the use of Java, JSP, and various web technologies. The programming phase includes the development of the system's server-side logic through Java Servlets, dynamic web content through JSP technology, a module for performing various numerical computations, a module for graphical visualization, a module for decision support through a rule-based system, as well as a module for managing classes and quiz assessments.

Deployment Phase - In this phase, the final product is deployed to a web server to enable users to access the **NumSolve 2.0** application through various desktop and mobile device browsers. After the application has been deployed, it is then monitored to provide any updates as needed to cater to the users' requirements.

The structured development process ensures that development progresses in a systematic manner, minimizes development risks, and aids in the success of the development of a dependable and user-friendly numerical analysis system..

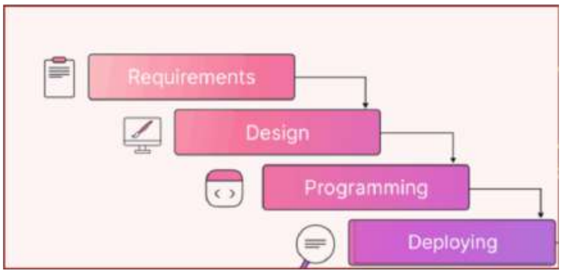


Figure 3.1: Waterfall Software Development Model

3.2.2 System Architecture: Three-Tier Design

The **Three-Tier Architecture** concept, which divides the system into three logical layers—the **Presentation Tier**, **Application Tier**, and **Database Tier**—is followed by the software architecture of **NumSolve 2.0**. Modularity, scalability, and maintainability are improved by this structure.

- **Presentation Tier:** This layer is the user interface layer of the system and is implemented using JSP, HTML, CSS, and Bootstrap. This is where user interactions such as entering an equation, entering parameters, and viewing numerical solutions and graphical solutions occur. Responsive web design techniques have been employed to make it mobile-friendly.

- **Application Tier:** The middle layer holds the system's processing logic. It provides the numerical algorithms and the decision support system based on rules and processes the user requests coming from the presentation layer. Java Servlets handle the communication between the web layer and the database layer.

- **Database Tier:** The lowest layer uses **MySQL** to control all data storage and retrieval operations. It stores user inputs, computation history, and generated reports by carrying out CRUD (Create, Read, Update, Delete) actions. Data security and consistency are preserved by this layer.

The system is easier to maintain or improve in the future because to its three-tier design, which also enhances code organization. Because each layer functions independently, developers can update one without having an impact on the others.

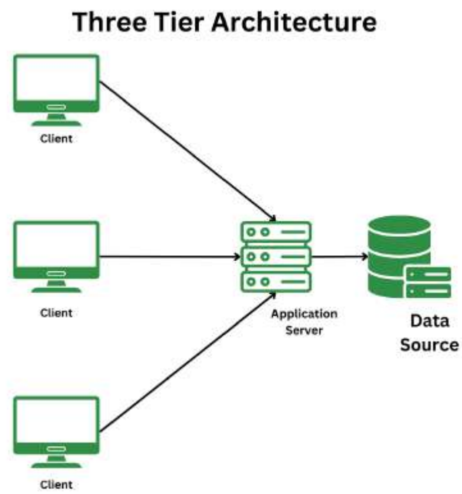


Figure 3.2: Three-Tier System Architecture of NumSolve 2.0

3.2.3 Software and Hardware Specification

The tables below outline the minimum software and hardware requirements for installing and running **NumSolve 2.0** efficiently.

Table 3.1: Software Specification

Item	**Specification**
Operating System	Windows / macOS / Linux
Programming Language	Java
Web Technologies	JSP, HTML, CSS, Bootstrap
Server	Apache Tomcat
Database	MySQL
Visualization Libraries	JavaScript Chart Libraries

Table 3.2: Hardware Specification

Item	**Specification**
Device	Desktop, Laptop, or Mobile Device
Processor	Dual-core processor or higher
Memory (RAM)	4 GB or higher
Internet Connection	Required for system access
Browser	Chrome, Firefox, Edge, or Safari

3.3 Project Planning Schedule

The project planning schedule provides a timeline of the development of NumSolve 2.0, ensuring that all research and technical objectives will be achieved in accordance with the timeline of the academic year. The project planning schedule coordinates the different phases of the Waterfall model with their respective deliverables and completion dates so that progress can consistently be tracked across the entire development life cycle. The detailed summary of the key milestones and the complete Gantt chart together serve as references for tracking tasks, allocating resources, and ensuring that the final system is delivered on time.

3.3.1 Milestone

The project milestone lists important phases and outputs for NumSolve 2.0's development. Every milestone in the project lifecycle signifies a major accomplishment or phase completion, enabling progress monitoring and guaranteeing that the project remains on track. Submission of paperwork, creation of prototypes, testing stages, and final presentations are examples of milestones. Setting these benchmarks guarantees consistency with the project goals and deadlines established by the FYP coordinator, aids in tracking progress, and helps maintain a clear timeline.

Table 3.3: Milestone for NumSolve 2.0

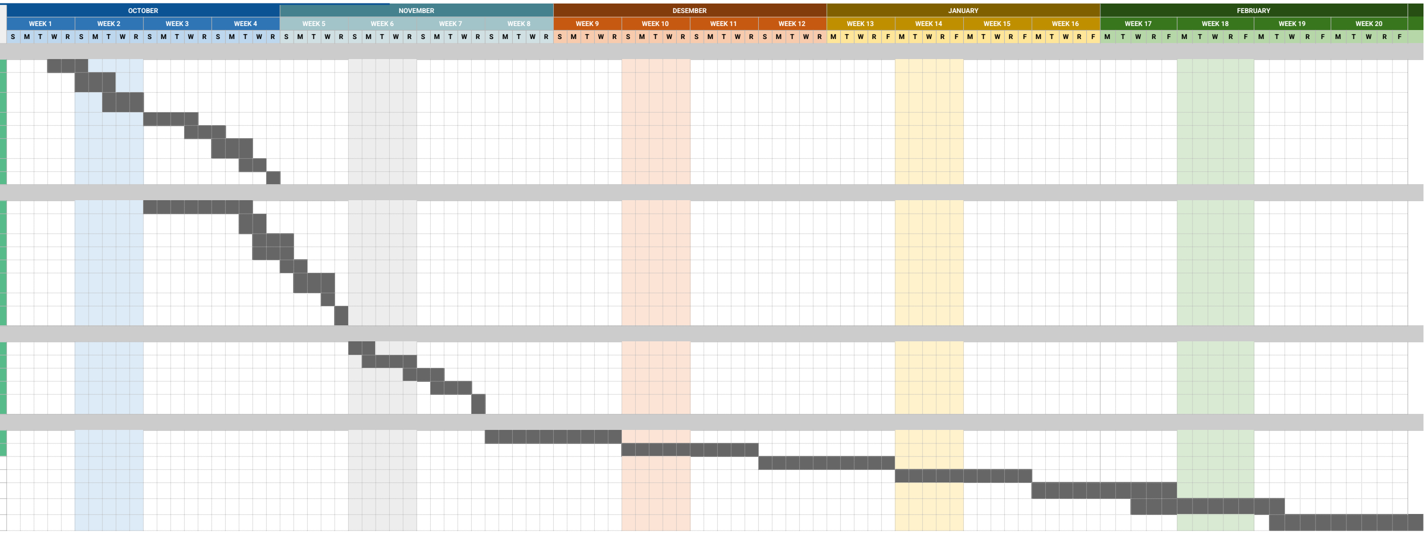
Milestone No.	Description	Start Date	End Date
1	Completion of Project Title	8 October 2025	9 October 2025
2	Completion of SPMP	15 October 2025	30 October 2025
3	Completion of SRS	22 October 2025	6 November 2025
4	Completion of SDD	29 October 2025	20 November 2025
5	Completion of Project Prototype	20 October 2026	9 June 2026
6	Completion of Thesis	20 March 2026	11 June 2026

3.3.2 Gantt Chart

The **NumSolve 2.0** project's overall development timetable is depicted by the Gantt chart, which details every stage from planning to implementation. To guarantee methodical advancement throughout the development process, it offers a clear visual depiction of project activities, task durations, and dependencies. The team can effectively allocate resources, keep an eye on milestones, and guarantee that each stage is completed on time in accordance with the project objectives by adhering to this planned timetable.

GANTT CHART

WBS NUMBER	TASK TITLE	START DATE	DUE DATE	DURATION	PCT OF TASK COMPLETE
1	**Planning**				
1.1	Approval of Project Title	08/10/25	12/10/25	4	100%
1.1.1	Define Problem and Solutions	12/10/25	14/10/25	2	100%
1.2	Define Objectives and Scopes	14/10/25	16/10/25	2	100%
1.3	Review on Related Systems	19/10/25	22/10/25	3	100%
1.4	Identify Methodology	22/10/25	26/10/25	4	100%
1.5	Identify Project Management Plan	26/10/25	28/10/25	2	100%
1.6	Complete SPMP Document	28/10/25	29/10/25	1	100%
1.7	Submission of SPMP Project	30/10/25	30/10/25	0	100%
2	**Requirements**				
2.1	Gather User Requirement	19/10/25	28/10/25	9	100%
2.2	Analysis The System Requirement	28/10/25	29/10/25	1	100%
2.3	Use Case Diagram	29/10/25	02/11/25	3	100%
2.4	Activity Diagram	29/10/25	02/11/25	3	100%
2.5	Class Diagram	02/11/25	03/11/25	1	100%
2.6	Sequence Diagram & CRUD Matrix	03/11/25	05/11/25	2	100%
2.7	Complete SRS Document	05/11/25	05/11/25	0	100%
2.8	Submission of SRS Document	06/11/25	06/11/25	0	100%
3	**Analysis & Design**				
3.1	Design System Architectural	09/11/25	10/11/25	1	100%
3.2	Design Database	10/11/25	13/11/25	3	100%
3.2.1	Design the User Interface	13/11/25	17/11/25	4	100%
3.2.2	Complete SDD Document	17/11/25	19/11/25	2	100%
3.3	Submission of SDD Document	20/11/25	20/11/25	0	100%
4	**Implementation**				
4.1	Implementation Module 1	23/11/25	04/12/25	11	100%
4.2	Implementation Module 2	07/12/25	18/12/25	11	100%
4.3	Implementation Module 3	21/12/25	01/01/26	10	0%
4.4	Implementation Module 4	04/01/26	15/01/26	11	0%
4.5	Implementation Module 5	18/01/26	29/01/26	11	0%
4.5	Implementation Module 6	27/01/26	08/02/26	11	0%
4.5	Implementation Module 7	09/02/26	20/02/26	11	0%



3.4 Summary

In conclusion, Chapter 2 has laid out a detailed description of how to structure the development of NumSolve 2.0 using the Waterfall Model so that there is a disciplined approach through all phases from gathering requirements through to deploying the system, while also ensuring that all levels of required mathematical accuracy necessary for performing numerical analysis are maintained. The implementation of Three Tier Architecture will help to deliver against the goals mentioned above by providing a framework that is modular, scalable and cross-platform and addressing the accessibility limitations in previous versions.

In addition, this chapter has described the necessary hardware and software specifications to ensure that the system is optimally configured for use in desktop and mobile environments. The inclusion of a project planning schedule and Gantt chart will provide guidance throughout the development lifecycle and identify key milestones such as the completion of the Rule Based Decision Module and database integration. Methodology and planning will provide the necessary foundation for all future phases of the project. With the methodology and planning established, research will move on to Chapter 4 which will examine the system's functional and non-functional specifications in detail.

CHAPTER 4

SYSTEM REQUIREMENTS

The introductory chapter will describe the basics of what will be required to implement the analytical models/forms of the NumSolve 2.0. It will start by discussing techniques used to elicit stakeholders' requirements and will then provide a more formal way to define both functional and non-functional requirements for the system. Finally, it will show how the system's architecture (including logic) may be broken down into three modeling forms (Use Case Diagram, Activity Diagram, Sequence Diagram) and provide a Class Diagram that models the static data structure. In addition, it will provide a CRUD Matrix that matches user roles and actions to data access opportunities, thereby creating an overall detailed development plan.

4.1 Introduction

To successfully implement NumSolve 2.0, we need to clearly transfer all the expectations of the users into the technical specifications. This chapter is the analytical core of the project, where we will document all of the essential functions and performance constraints that are necessary to develop a successful numerical computation platform. To provide a robust structure for the design and coding phases, we used standard software engineering models (behavioral activity flows, structural class relationships, etc.) to create an appropriate framework. The end result will be a system that meets the educational and computational needs of both students and educators.

4.2 Requirement Elicitation Techniques

The process of obtaining data from stakeholders in order to specify the limitations and functionalities of the system is known as requirement elicitation. In order to make sure the finished system achieves its goals, it aims to determine user needs and expectations. Discussions, observations, and document analysis with students, teachers, and the system analyst were all part of the NumSolve 2.0 process.

4.2.1 Requirement Sources

The requirements for the **NumSolve 2.0** system were compiled from primary sources, which are listed in this section. Comprehending these sources guarantees that the final system takes into account the technical and educational requirements of its stakeholders. Three primary sources of information were gathered: reference materials, current systems, and stakeholders.

Stakeholders

The main stakeholders involved in this project are as follows:

- **Students:** represent the primary users of the system who require an easy-to-use and efficient tool to carry out and comprehend numerical computations. They look for assistance in choosing suitable numerical techniques, interactive learning aids, and graphical visualization.
- **Educators:** Incorporate researchers and instructors who use the system for instruction and analysis. In order to improve study analysis and teaching efficacy, they place a strong emphasis on correctness, visualization support, and theoretical justifications.
- **System Analyst:** In charge of preserving system dependability, keeping an eye on performance, examining usage data, and controlling user access. Throughout upgrades, the analyst makes that the system stays effective, safe, and functionally consistent.

Documents

To find key algorithms, computational strategies, and theoretical ideas, documents including university course materials and textbooks on numerical analysis were examined. To guarantee academic and technical accuracy, these references served as a guide for the creation and validation of the numerical techniques used in **NumSolve 2.0**.

Existing Systems

To determine their advantages and disadvantages, a number of software programs were examined, including MATLAB, Wolfram Alpha, GeoGebra, Scilab, and the most recent version of NumSolve 1.0. Even while these tools offer sophisticated symbolic and numerical computation skills, they frequently call for a lot of system resources or constant internet access. In order to overcome these drawbacks, **NumSolve 2.0** offers a web-based and platform-independent solution that integrates intelligent learning support, numerical computation, and visualization into a single browser-accessible system.

4.2.2 Requirement Techniques

The methods for collecting and evaluating the requirements for the creation of the **NumSolve 2.0** system are described in this section. The methods chosen guarantee that, while preserving technical accuracy based on recognized scholarly sources, the data gathered appropriately represents the expectations and difficulties of the intended users. Conversational, observational, and document-centric approaches were the three main methods employed.

Conversation Techniques

To determine their demands and challenges in carrying out numerical computations, interviews and casual conversations were held with a few math students and instructors, including senior lecturers in numerical analysis. Students discussed their difficulties using current tools, including MATLAB, which frequently need extremely complex setup or previous programming experience. Teachers highlighted the value of clear theoretical explanations and visual aids to enhance instruction. The data acquired from these discussions served as a guide for identifying the essential components of the system, such as the graphical visualization, learning support, and rule-based recommendation module.

Observation Techniques

Existing numerical computation platforms, including MATLAB, Wolfram Alpha, GeoGebra, Symbolab, and the previous iteration of the system, NumSolve 1.0, were analyzed during observation sessions. Examining their computational procedures, interface design, and usability was the aim. This finding led to the identification of a number of problems, such as the setup's complexity, reliance on internet access, and the computation parameters' restricted control. In particular, NumSolve 1.0 had few visualization features and no intelligent recommendations. The creation of NumSolve 2.0 was directed by these conclusions in order to guarantee a more intuitive user interface, flexible parameter control, and improved learning support through a web-based platform.

Document-Centric Techniques

To make sure that all of the numerical techniques used in **NumSolve 2.0**—such as bisection, secant, Newton-Raphson, interpolation, differentiation, and integration—follow accurate and standardized mathematical formulas, reference materials, including textbooks and course notes on numerical analysis, were examined. This method allowed the system to function as a computational and instructional tool while guaranteeing academic accuracy and consistency.

4.3 System Requirements

NumSolve 2.0's full set of features and performance requirements are outlined in the System Requirements section. It outlines the functional requirements (what the system must accomplish) and the non-functional requirements (how well the system must fulfill those activities). In order to ensure that all identified user and business needs are converted into precise, workable specifications, this part serves as a link between the requirement elicitation process and the design phase. The design, implementation, and validation of the system are based on the requirements listed here.

4.3.1 Functional Requirement

Functional requirements outline the essential functions and features that the **NumSolve 2.0** system must have in order to satisfy project and user goals. They detail the roles and interactions of the system's users, namely the **Students**, **Educators**, and the **System Analyst**, and specify what the system should do in particular scenarios.

Every functional requirement is made to guarantee dependability, usability, and accessibility while executing numerical calculations, displaying outcomes, and assisting with academic learning.

Student

The system's main users, who carry out numerical calculations and use it as a learning aid, are students. The following features must be available to students through the system:

- **FR1:** The system shall allow students to register, log in, and manage their profiles securely.
- **FR2:** The system shall allow students to input equations or datasets for numerical computations.
- **FR3:** The system shall perform computations using multiple numerical methods, including root-finding, interpolation, differentiation, integration, and ODE solvers.
- **FR4:** The system shall provide a rule-based decision module to recommend the most suitable numerical method based on user input.
- **FR5:** The system shall display graphical visualizations and step-by-step iterations of the selected numerical methods.
- **FR6:** The system shall allow students to save, view, update, and delete their computation history.
- **FR7:** The system shall allow students to generate and export computation results and reports in PDF or CSV format for reference or submission.
- **FR8:** The system shall allow students to view and access learning materials, including notes, tutorials, worked examples, and multimedia content.
- **FR9:** The system shall allow students to participate in quizzes related to numerical methods and computations.
- **FR10:** The system shall provide gamification elements such as scores, badges, or progress levels to motivate learning.
- **FR11:** The system shall allow students to join and view class groups created by educators.

Educator

Researchers and lecturers who use the method to instruct, mentor, and assess students' comprehension of numerical analysis are considered educators. The following features must be included in the system for educators:

- **FR12:** The system shall allow educators to upload new learning materials, including notes, tutorials, worked examples, or multimedia content.
- **FR13:** The system shall allow educators to update existing learning materials, including file content, title, description, or metadata.
- **FR14:** The system shall allow educators to delete outdated or incorrect learning materials.
- **FR15:** The system shall allow educators to view and access learning materials, just like students.
- **FR16:** The system shall allow educators to create and manage quizzes for students.
- **FR17:** The system shall allow educators to configure gamification rules such as scoring and rewards.
- **FR18:** The system shall allow educators to create, manage, and assign students to classes.

System Analyst

To guarantee stability and dependability, the System Analyst is in charge of maintaining and keeping an eye on the **NumSolve 2.0** system's overall performance. The following features must be included in the system for the system analyst:

- FR19: The system shall allow the system analyst to view system activity logs, user statistics, and performance reports.
- FR20: The system shall allow the system analyst to manage user accounts and access permissions.
- FR21: The system shall allow the system analyst to monitor quiz activity, class usage, and gamification performance.
- FR22: The system shall allow the system analyst to configure and maintain system-wide class and quiz settings.

Summary

Together, these functional requirements make sure that **NumSolve 2.0** assists its intended users, allowing **students** to learn and compute, **educators** to mentor and teach, and the **system analyst** to effectively monitor and maintain system operations.

4.3.2 Non-functional Requirement

A collection of specifications known as non-functional requirements (NFRs) outline the **NumSolve 2.0** system's operating characteristics, performance benchmarks, and environmental limitations. These specifications place more emphasis on the system's ability to carry out its intended tasks than on its actual capabilities. They are necessary to maintain high user satisfaction and long-term maintainability while ensuring the program runs effectively, securely, and dependably.

Performance, scalability, portability, compatibility, stability, availability, maintainability, security, localization, and usability are among the most frequently cited non-functional criteria. Each of these prerequisites enhances the system's overall quality and guarantees that **NumSolve 2.0** may be utilized successfully in a range of computational and academic settings.

- **NFR1: Performance and Scalability**

  The system should be able to perform basic numerical computations like root solving, differentiation, and interpolation in **3 seconds** for most input sizes. More complex numerical computations like solving an ODE or large data sets should take **5 seconds**. The graphics should render in **2 seconds** once the computation is finished. The architecture should be scalable, so new numerical services or new users can be added without affecting the performance of the system.

- **NFR2: Portability and Compatibility**

  The system shall be developed in the form of **web-based software** that can be accessed using common web browsers like Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari. The system shall be compatible with common operating systems like Windows, macOS, and Linux. The system shall run without any need to install or rely on any external computational platform like MATLAB or Python.

- **NFR3: Reliability, Maintainability, and Availability**

  The system shall function in a reliable manner and to a large extent without failure and shall have at least **99%** uptime during normal academic use times. The system shall ensure that all the results obtained in numbers shall be similar and shall be reproducible when given similar parameters. The system source code shall ensure that there is ease of understanding and modification.

- **NFR4: Security**

  The system will employ strong authentication and authorization. This will ensure that the accounts and data of the users are secure. The passwords of the users will be protected using secure hashing techniques. The sensitive data will be protected from unauthorized access, modification, or deletion. The input validation and proper session management will be implemented to overcome the common security threats.

- **NFR5: Localization and Compatibility with Local Context**

  The system is set to use English in its interface. However, in the future, other languages may be supported. The numbers, reports, and data exports will all be in accordance with global academic standards to ensure compatibility between the regions and the global aspects in education and research.

- **NFR6: Usability and User Interface Design**

  The system's interface must be intuitive and user-friendly, whether you're a beginner or a long-term user. Users should be able to do everyday activities, for example, typing out equations and calculating results with a minimal instruction manual. This is made possible by good layout, the assistance of tooltips, and validation information. Lastly, the user interface must be created through the usage of JSP, HTML, and CSS, and must be responsive so that the system runs properly on different displays.

- **NFR7: Data Integrity and Accuracy**

  The calculations being performed should be done through tried-and-tested algorithms to ensure accuracy of results. Data consistency throughout the calculation, storage of the data, and creation of the report needs to be maintained by the system. Error handling needs to be implemented to ensure that no damage to data occurs. No incorrect outcomes of calculation should also occur.

These non-functional requirements guarantee that **NumSolve 2.0** functions as a reliable, secure, and effective learning platform that satisfies contemporary academic and usability standards in addition to being a useful and intelligent computational tool.

4.4 Requirement Analysis

In this part, we describe a detailed logical model of the system through the use of Unified Modeling Language (UML). In section 4.3, we received functional requirements for the system, which we used to create visual workflows and structural designs that assist in defining user system interactions and establishing accurate definitions for all data relationships prior to implementation via requirement analysis.

4.4.1 Use Case Diagram

The Use Case Diagram is an overview of high-level functionality within the system, providing an illustration of the actors (i.e., Students, Educators, and System Analysts) and their corresponding goal(s). From one diagram, you can see at a high level the whole system scope/what functionalities exist for each user role.

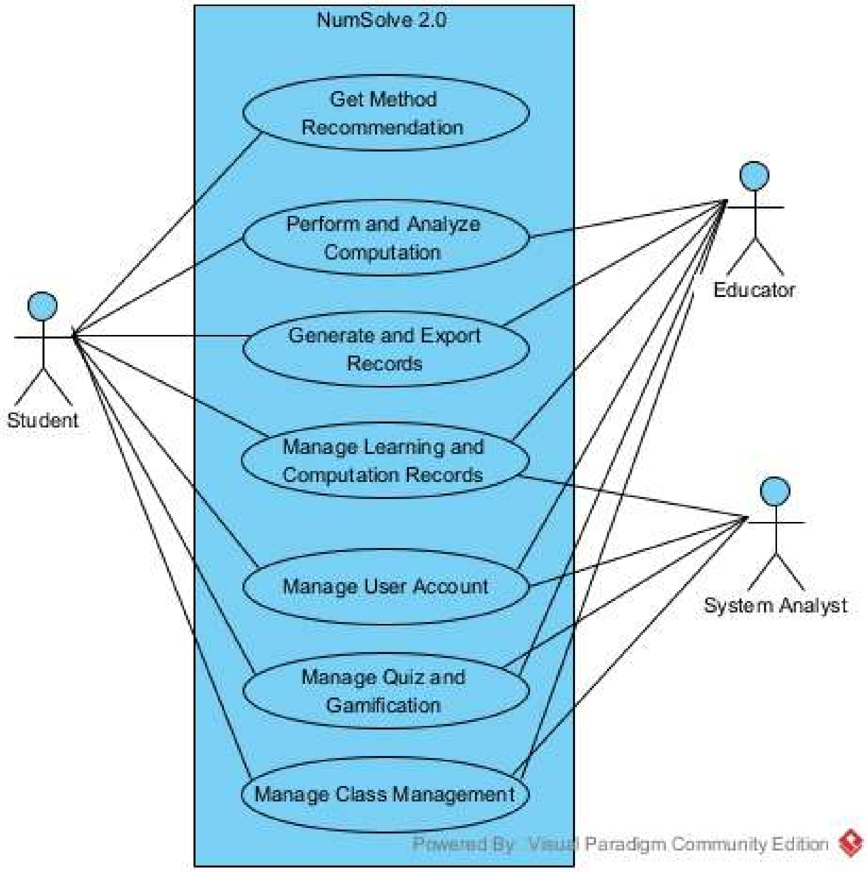


Figure 4.1: Use Case Diagram of Numsolve 2.0

4.4.2 Use Case Descriptions

Table 4.1: Use Case 1: Manage User Account

Use Case Name:	Manage User Account	**ID:** UC1
Importance Level:	High	
Primary Actor:	Student, Educator, System Analyst	**Use Case Type:** Primary, Essential
Stakeholder and Interests: Students and educators manage personal accounts. System Analyst ensures account integrity and security.		
Brief Description: Users can register, log in, and update account details. The System Analyst manages and maintains user accounts.		
Normal Flow of Events: 1. User access the system through web browser. 2. System displays authentication page. 3. User enters credentials or registration info. 4. System validates and grants access or creates account. 5. User updates profile if needed. 6. System Analyst may manage, modify, or delete accounts.		
Alternate/Exceptional Flows: 1a. Invalid credentials prompt an error and reentry.		

Table 4.2: Use Case 2: Get Method Recommendation

Use Case Name:	Get Method Recommendation	**ID:** UC2
Importance Level:	High	
Primary Actor:	Student, Educator	**Use Case Type:** Primary, Essential
Stakeholder and Interests: Users want guidance on selecting the most suitable numerical method for their problem.		
Brief Description: System analyzes input equations or datasets and suggests optimal numerical methods with explanations.		
Normal Flow of Events: 1. User selects recommendation option. 2. Input is validated by the system. 3. Rule-based module analyzes the problem. 4. Recommended method(s) are displayed with rationale.		
Alternate/Exceptional Flows: 2a. Invalid input format triggers an error and requests correction.		

Table 4.3: Use Case 3: Perform and Analyze Computation

Use Case Name:	Perform and Analyze Computation	**ID:** UC3
Importance Level:	High	
Primary Actor:	Student, Educator	**Use Case Type:** Primary, Essential
Stakeholder and Interests: Users perform calculations using selected numerical methods and view detailed results.		
Brief Description: Run numerical computations and display outputs with graphs and stepwise breakdown.		
Normal Flow of Events: 1. User selects numerical method. 2. Input parameters are provided. 3. System executes computation. 4. Results and graphs are displayed. 5. User may save or export results.		
Alternate/Exceptional Flows: 3a. Invalid inputs cause error messages with correction suggestions.		

Table 4.4: Use Case 4: Manage Learning and Computation Records

Use Case Name:	Manage Learning and Computation Records	**ID:** UC4
Importance Level:	Medium	
Primary Actor:	Student, Educator, System Analyst	**Use Case Type:** Primary, Essential
Stakeholder and Interests: Users manage previous computations and teaching resources. System Analyst ensures database integrity.		
Brief Description: Create, update, or delete stored computations and learning materials.		
Normal Flow of Events: 1. User opens records section. 2. System displays available materials. 3. User selects record to view or edit. 4. Educator may upload or update materials. 5. System updates database.		
Alternate/Exceptional Flows: 4a. File upload failure triggers retry prompt.		

Table 4.5: Use Case 5: Generate and Export Records

Use Case Name:	Generate and Export Records	**ID:** UC5
Importance Level:	Medium	
Primary Actor:	Student, Educator, System Analyst	**Use Case Type:** Primary, Essential
Stakeholder and Interests: Students and educators generate computation records for learning, submission, or analysis purposes, while the system analyst generates summary records for monitoring and evaluation.		
Brief Description: Allows users to generate, view, and export computation and learning records in PDF or CSV format.		
Normal Flow of Events: 1. User navigates to the records or reports section. 2. User selects a computation or learning record. 3. System compiles the selected data. 4. User chooses the desired export format. 5. System generates and saves the exported file.		
Alternate/Exceptional Flows: 5a. If data retrieval fails, the system displays an error message and prompts the user to retry.		

Table 4.6: Use Case 6: Manage Quiz and Gamification

Use Case Name:	Manage Quiz and Gamification	**ID:** UC6
Importance Level:	Medium	
Primary Actor:	Student, Educator	**Use Case Type:** Primary, Essential
Stakeholder and Interests: Educators require tools to assess student understanding through quizzes, while students benefit from gamification features that enhance motivation and engagement.		
Brief Description: Allows educators to create and manage quizzes and gamification settings, while students participate and track progress.		
Normal Flow of Events: 1. Educator creates or updates a quiz. 2. System stores quiz configuration. 3. Student attempts the quiz. 4. System evaluates submitted answers. 5. System updates scores, badges, or progress indicators.		
Alternate/Exceptional Flows: 6a. Invalid quiz configuration or submission triggers an error message and correction prompt.		

Table 4.7: Use Case 7: Manage Class Management

Use Case Name:	Manage Class Management	**ID:** UC7
Importance Level:	Medium	
Primary Actor:	Educator, Student, System Analyst	**Use Case Type:** Primary, Essential
Stakeholder and Interests: Educators require structured class management; students need access to classes; system analyst ensures data integrity.		
Brief Description: Enables educators to create classes, students to join, and analysts to oversee structures.		
Normal Flow of Events: 1. Educator creates a new class. 2. Students join or are assigned to the class. 3. System stores and updates class information. 4. Users view class-related content and records.		
Alternate/Exceptional Flows: 7a. Duplicate class name or invalid enrollment triggers an error message and retry request.		

4.4.3 Activity Diagrams

n addition to defining what the system does, the Activity Diagrams model the internal logic and sequence of the tasks. The following diagrams illustrate the step-by-step decision-making processes for the system's core functionalities such as the recommendation engine and report generation, focusing on both successful and erroneous paths through the system.

Activity Diagram 1: Manage User Account

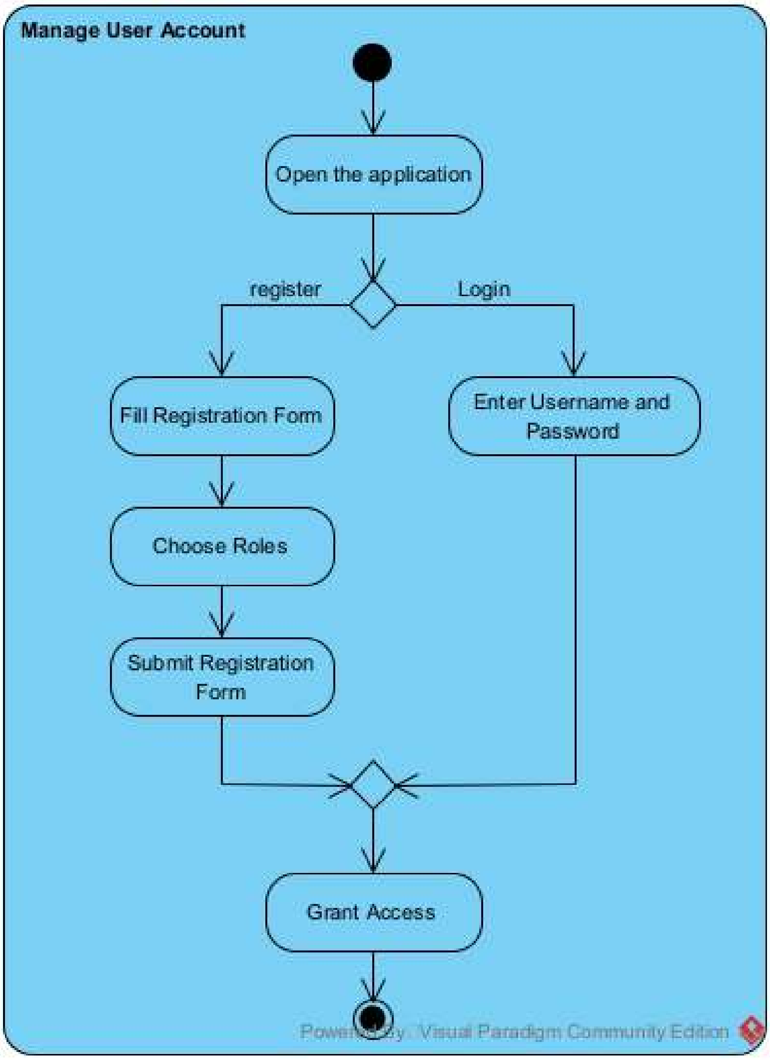


Figure 4.2: Activity Diagram for Manage User Account

Description: This diagram outlines how users interact with the system to create, access, or manage their personal accounts in NumSolve 2.0. The process begins when the user access the system through web browser and the user selects either login if have entered the app before or registration if first time using the app. The system checks the entered details and determines whether access should be permitted or not.

Activity Diagram 2: Get Method Recommendation

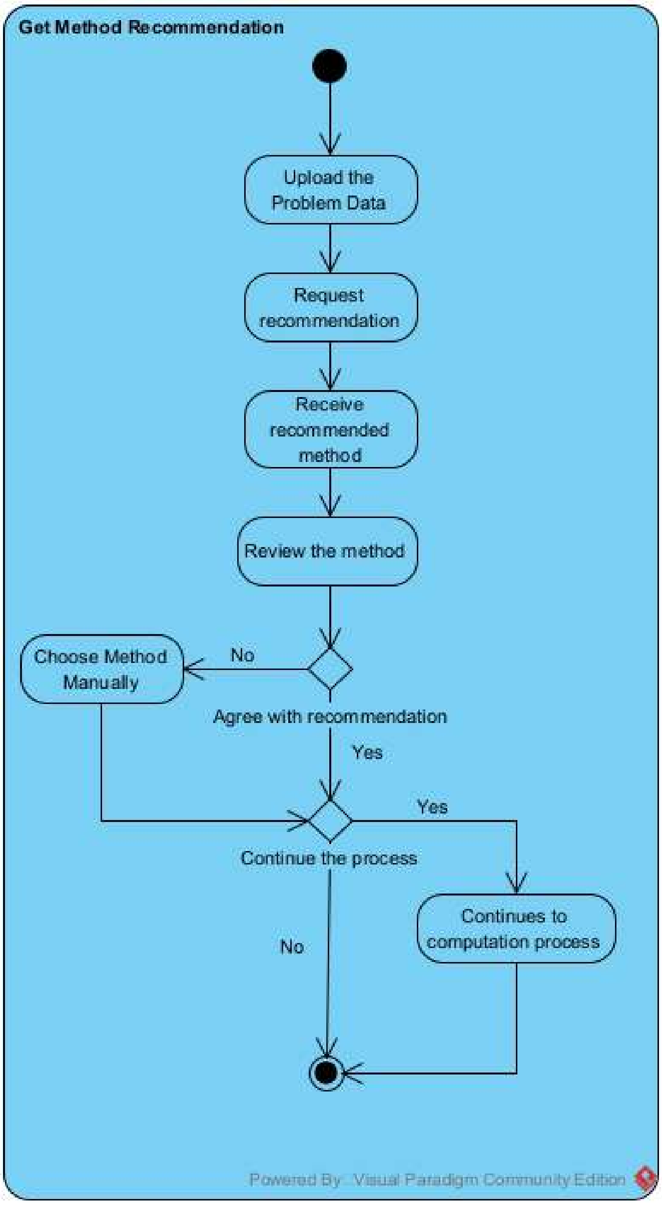


Figure 4.3: Activity Diagram for Get Method Recommendation

Description: This Diagram show how system provide guidance to choose the best method. The user will submit or upload the equation which will be evaluated by the system. After been evaluated, the system will recommend the best method to be used and we can choose to follow the recommendation or not. If no, then we can choose the method manually but if we agree then we can continue the computation process or not.

Activity Diagram 3: Perform and Analyze Computation

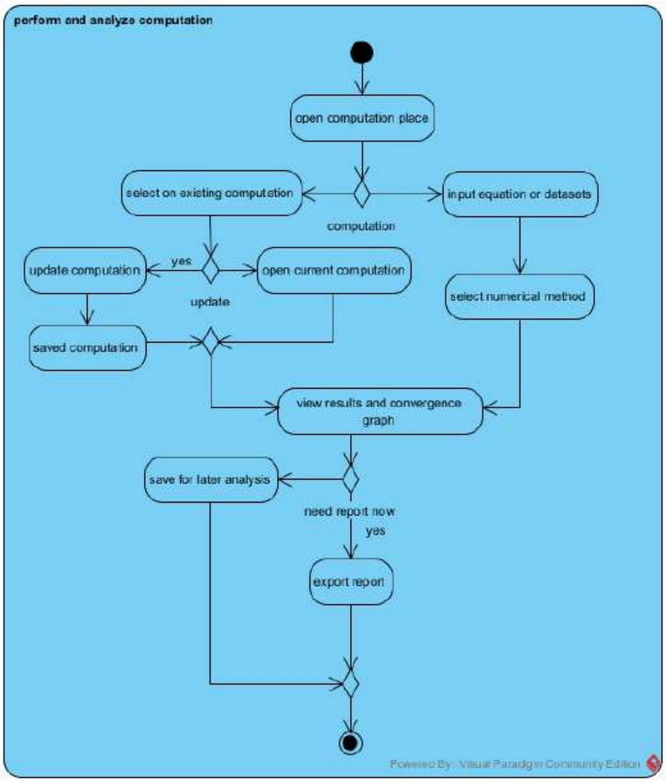


Figure 4.4: Activity Diagram for Perform and Analyze Computation

Description: The diagram show the Computation process. The user firstly must enter equation or datasets or they can open the computation that have been made. For both they can update if want to change the recorded one or to make new one. Then after that they must choose the numerical method that they want to use. After that, the system will calculate and review the answer and convergence graph. The user can choose for detail information or not. If yes then it will show the step by step iterations to get extra knowledge. The user also can export the report or if dont want to export immedietly then they can save for later analysis use.

Activity Diagram 4: Manage Learning Materials

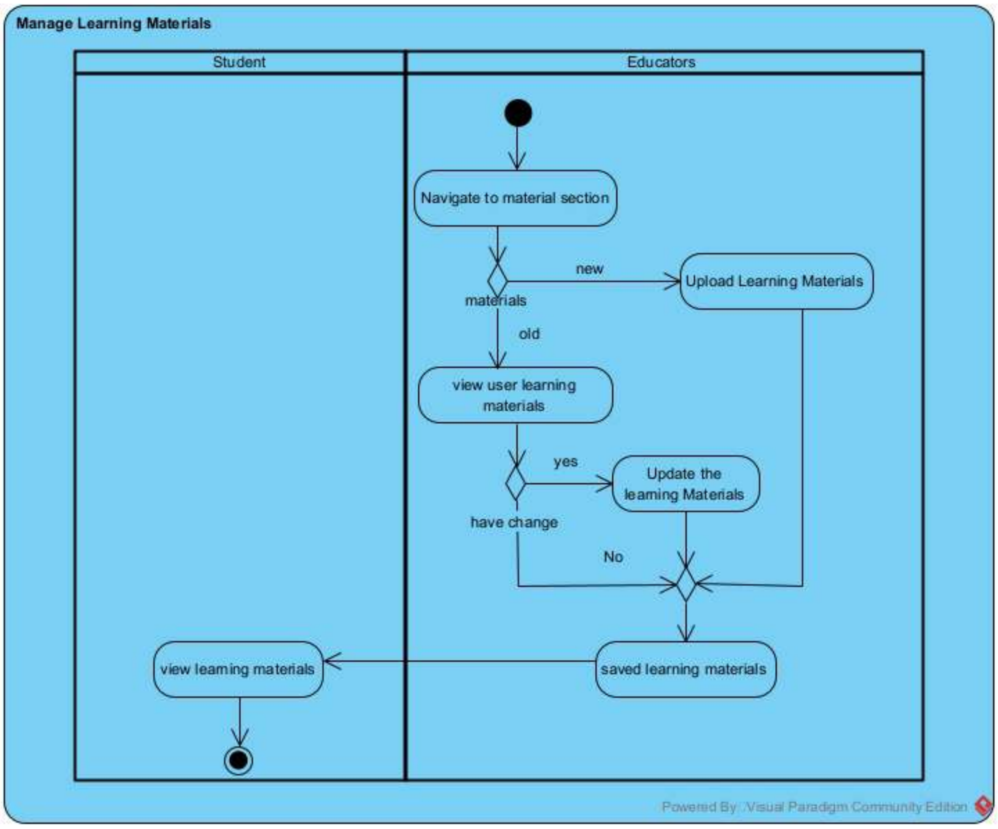


Figure 4.5: Activity Diagram for Manage Learning Materials

Description: This diagram show workflow for managing learning materials. Educators and student can open the materials section. For student they can only search the materials that they want and the system will show the materials they wanted. For Educators they also can search the materials and they can create their own materials. If Educators want, they can upload their materials to the system and saved it or they can update their materials also.

Activity Diagram 5: Generate and Export Reports

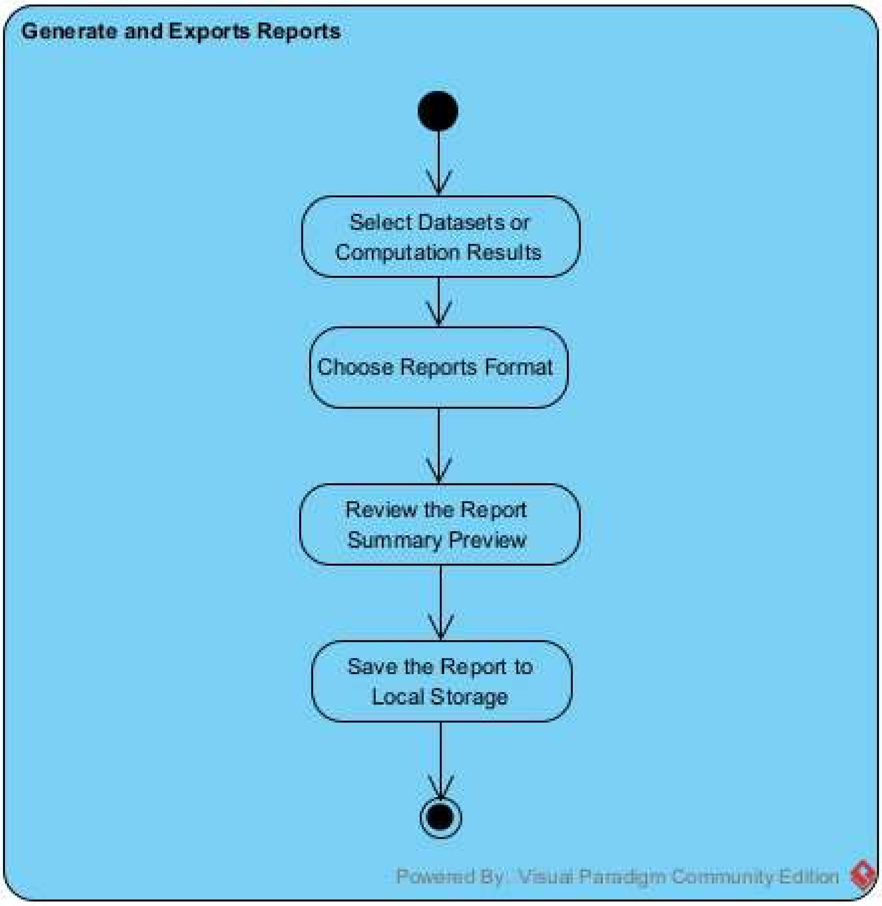


Figure 4.6: Activity Diagram for Generate and Export Reports

Description: This diagram shows the process of generate and export reports. The user can select datasets or computation records that they want to export. Then the user need to choose type of report format between CSV and PDF. Before exporting they can review the report summary to make sure they approved it. Then after approved it, they can saved the report to the local storage on their gadget.

Activity Diagram 6: Manage Quiz and Gamification

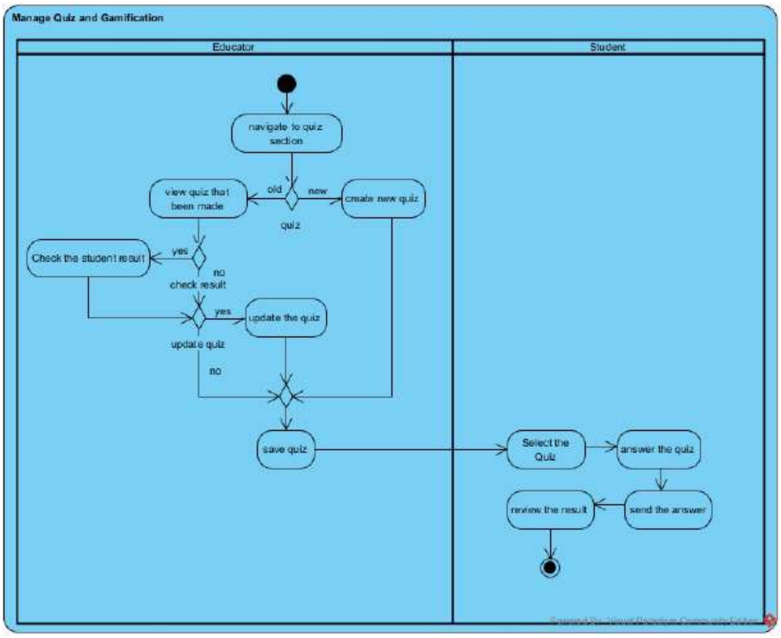


Figure 4.7: Activity Diagram for Manage Quiz and Gamification

Description: This diagram illustrates the management and attempt of the quizzes. The Educator will start by looking at the list of quizzes. The Educator can then choose either to create a new quiz or edit an existing one. The Educator can then edit the questions or settings and even look at the scores of the students. The Educator then clicks 'save' after finishing. The Student will then start by selecting the title of the quiz. The Student will then attempt the questions and click 'submit' after finishing. The system will then calculate and display the result with a badge for the Student.

Activity Diagram 7: Manage Class management

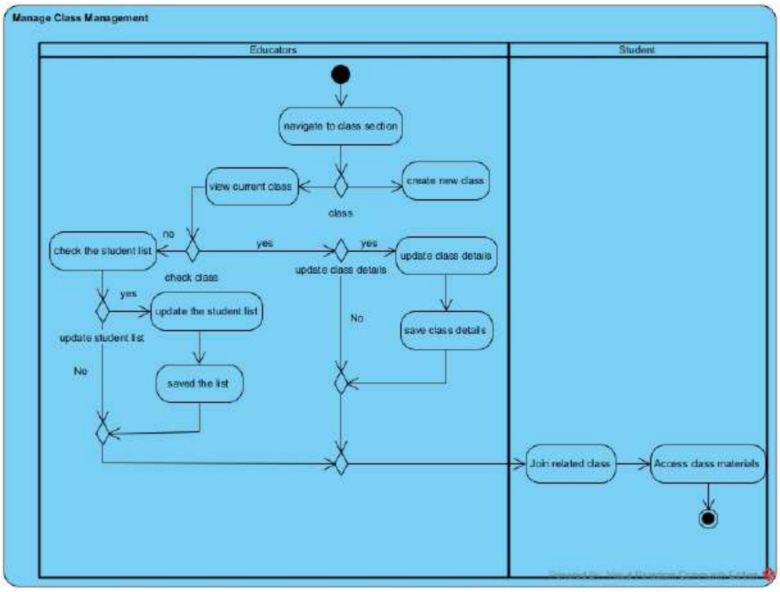


Figure 4.8: Activity Diagram for Manage Class management

Description: The diagram above illustrates the process of class info management, which includes student enrollment in the class by the Educator. The Educator starts by viewing the classes currently available, which then gives the option to create a new class or edit the details of an existing class. Within the edit class details option, the Educator can manage the class by assigning new students to the class or removing students who are currently assigned to the class. After completing the above options, the class info is then saved by the Educator. The role of the Student is to request to be assigned to a class, which then gives the student the option to view the class details.

4.4.4 Class Diagram

The class diagram for NumSolve 2.0 offers a static perspective to the system, including the key classes, their properties, methods, and interactions between the various classes. The diagram represents how students, teachers, and system analysts communicate with other components including User Account, Computation, Method Recommendation, Records, and Reports. The class diagram assists in appreciating the system structure as well as implementing the system functions.

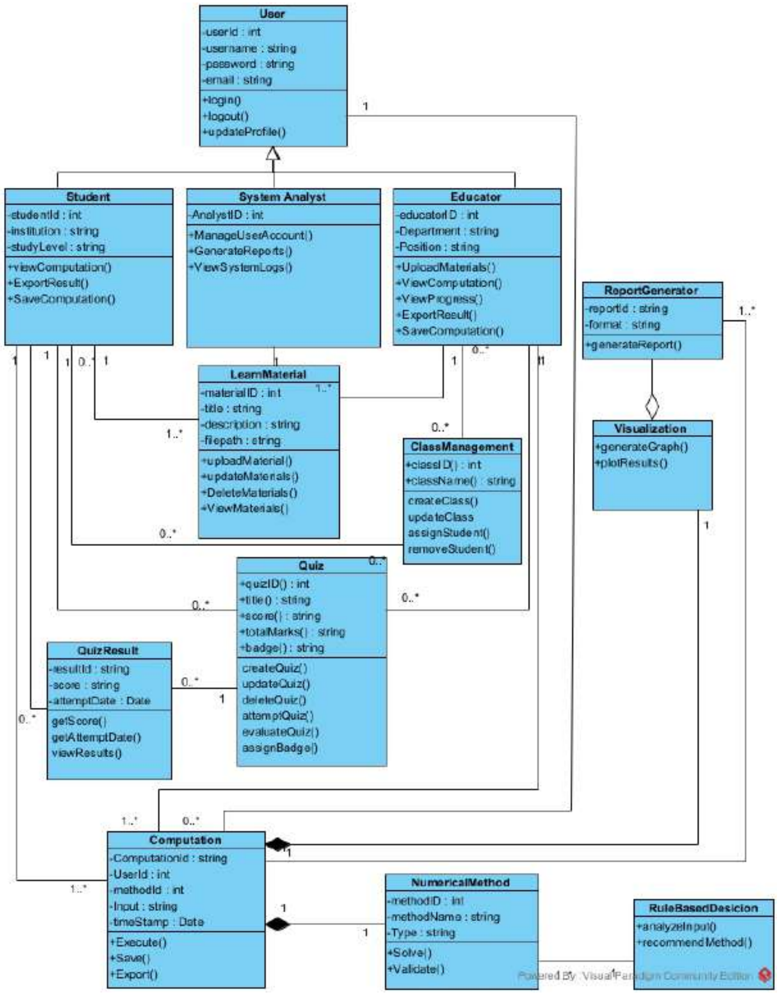


Figure 4.9: Class Diagram of NumSolve 2.0

Description: The class diagram above gives an overview of the system structure as well as the relationships between the main components of the system. The main class in this system is the User class, which contains basic information needed to identify users. This class is then extended into three roles: Student, Educator, and System Analyst.

The Student class enables participation in quizzes, viewing of computational results, saving of results, and keeping track of student records. The Educator class oversees learning activities that include class creation, class management, learning material upload, quizzes, and student tracking. The System Analyst class oversees activities such as account management, report generation, and log files.

With the following classes, the educational materials are managed: ClassManagement, LearnMaterial, and Quiz. LearnMaterial takes care of the learning materials, ClassManagement deals with class creation and assignment of students to classes, and Quiz takes care of quiz creation and evaluation using QuizResult.

The role of the Computation class is to handle all the computational processes. It will store all the inputted data, method selection, and computation results. The Computation class interacts mainly with the NumericalMethod class, which describes the methods for solving mathematical equations, and the RuleBasedDecision class, which recommends the best method based on user input.

Finally, the ReportGenerator and Visualization classes provide structured reporting and a graphical representation of data, enabling easy interpretation of computational and performance data.Overall, the above diagram clearly indicates a comprehensive view of user management, educational resources, computation, and reporting within the system.

4.4.5 Sequence Diagrams

Sequence diagrams for NumSolve 2.0 illustrate the interactions between actors and the system for each main functionality. They show the sequence of messages exchanged between objects, highlighting the order of operations, decision points, and system responses. These diagrams complement the use-case and activity diagrams by providing a dynamic view of the system behavior during each process.

Sequence Diagram 1: Manage User Account

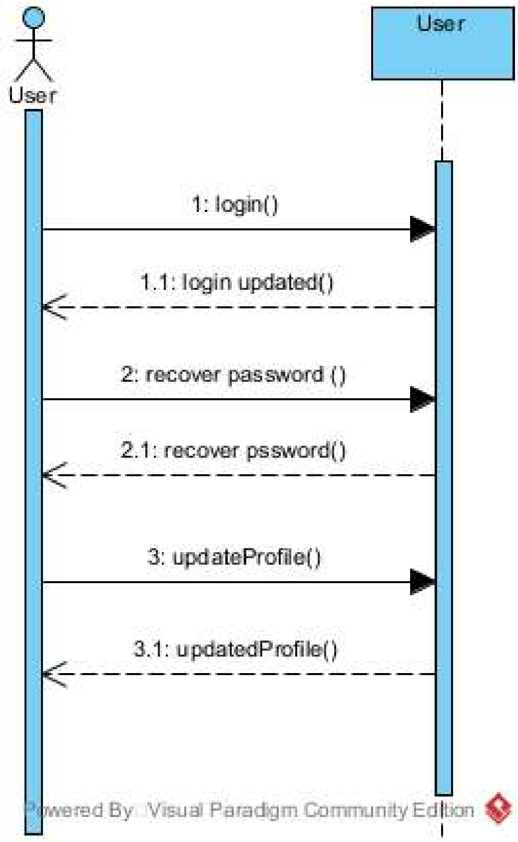


Figure 4.10: Sequence Diagram for Manage User Account

Description: diagram represents the interaction of a user, either a student, educator, or system analyst, with the system that manages the user's account. The user first opens the system, adds personal details, a role, and submits the registration form. The system then checks that the user has registered by confirming that the registration is complete. The user can then log in to the system by entering a password and a user name. The user can forget the password, so they can recover it by executing the recoverPassword() function. The user can also change some of the personal details by executing the updateProfile() function, upon which the system updates the profile in the account.

Sequence Diagram 2: Get Method Recommendation

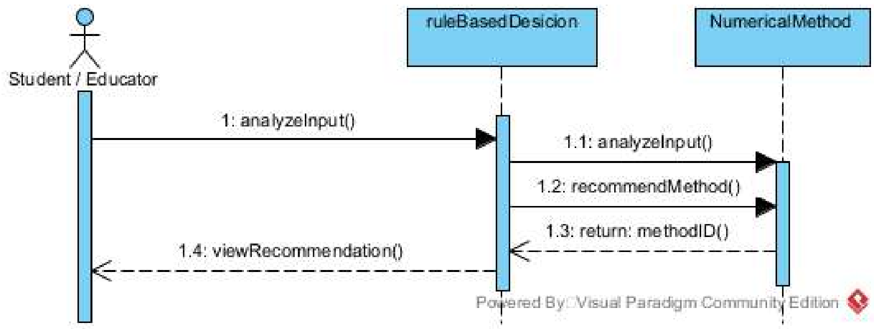


Figure 4.11: Sequence Diagram for Get Method Recommendation

Description: The following sequence diagram illustrates the process involved in the Rule Based Decision System. The Student or Educator initiates the process by sending a query to analyze the equation string using the analyzeInput() function. The RuleBasedDecision object examines this query and consults the NumericalMethod class to examine the properties of the equation. The most appropriate numerical method is determined using the recommendMethod() function, and the Method ID is returned to the user.

Sequence Diagram 3: Perform and Analyze Computation

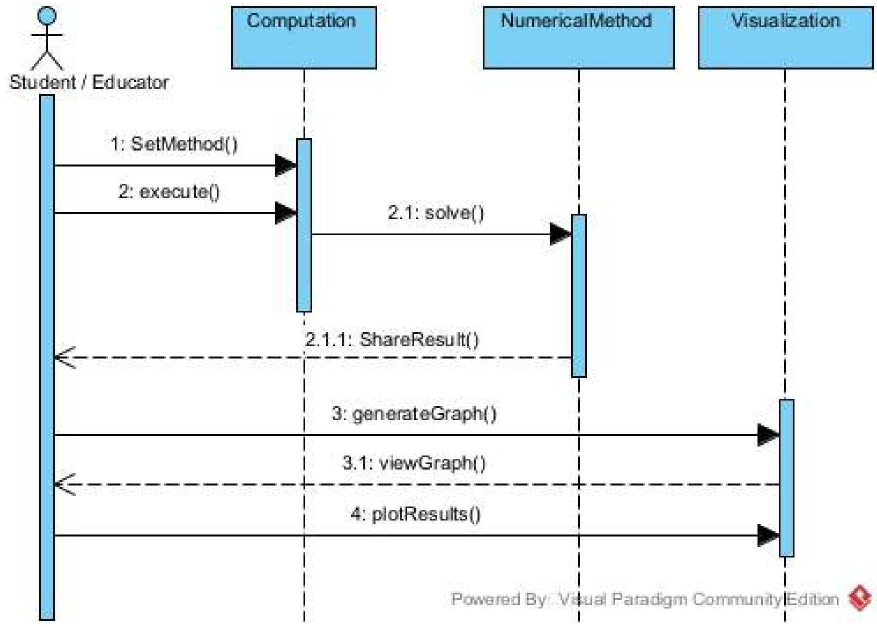


Figure 4.12: Sequence Diagram for Perform and Analyze Computation

Description: This is a description of a process of performing a numerical computation. As shown in the figure, a user is first required to input a method ID and the appropriate input data. At this point, the Computation object initiates the Execute command, which in turn invokes the Solve method of the class named NumericalMethod. After computation, the user is then allowed to view the results. Using the class named Visualization, the user is able to create a graph of the results, as well as a convergence chart, using the methods named generateGraph and plotResults, respectively.

Sequence Diagram 4: Manage Learning Materials

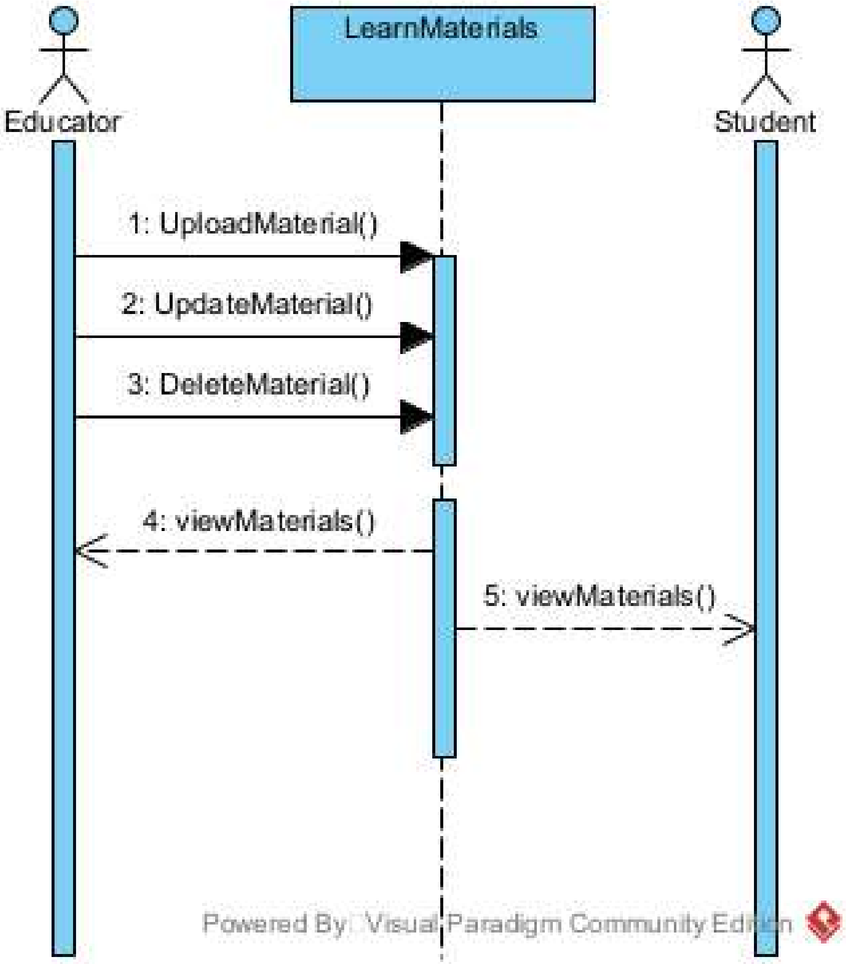


Figure 4.13: Sequence Diagram for Manage Learning materials

Description: This is a sequence diagram showing management of educational materials/content. Here, an Educator sends an uploadMaterial() or updateMaterials() request in order to upload new materials or update existing ones, respectively. This is then acknowledged by the system. There is also an action where an Educator can delete materials/content using the DeleteMaterials() function. Both an Educator and Student can access materials/content, as depicted in this diagram where they both request ViewMaterials().

Sequence Diagram 5: Generate and Export Reports

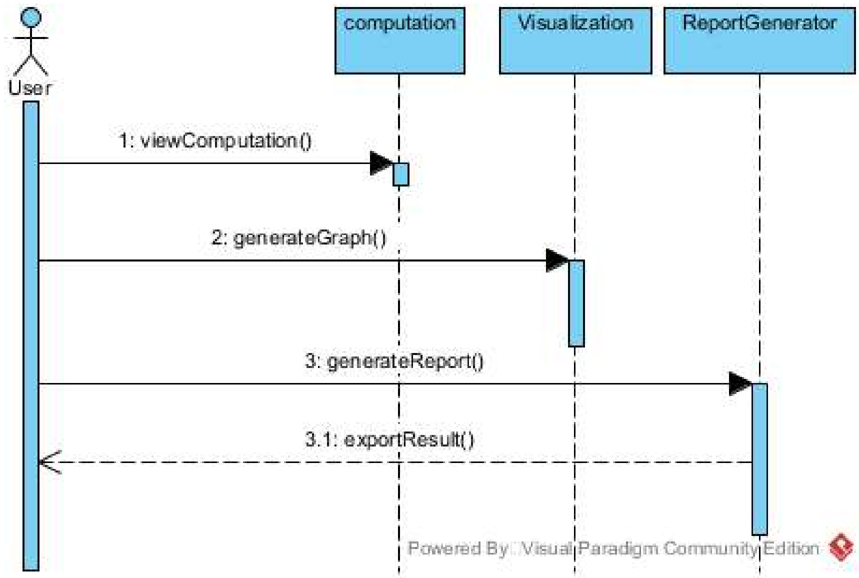


Figure 4.14: Sequence Diagram for Generate and Export Reports

Description: This is a diagram representing the process involved in generating and exporting reports from saved data. A user begins by retrieving a saved data record from the system using the viewComputation method. They then proceed to request a visual summary, thus activating the generateGraph method in the Visualization class. To obtain a formal report, the user is required to specify their preferred format before proceeding to request the generation of a report through the ReportGenerator class. They then proceed to request an export of the data in their preferred format through the ExportResult method, after which it is exported by the Computation class.

Sequence Diagram 6: Manage Quiz and Gamification

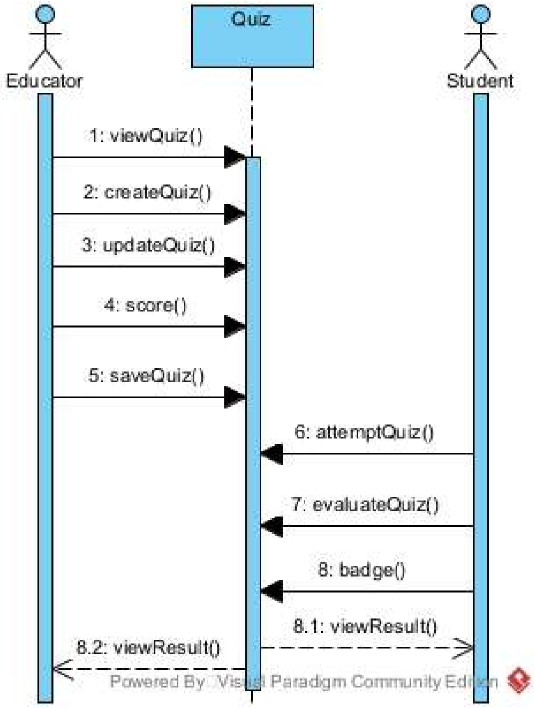


Figure 4.15: Sequence Diagram for Manage Quiz and Gamification

Description: illustrates the interactions for quiz management and participation, which are performed by the Educator and the Student, respectively. The Educator will first invoke the function viewQuiz() to display the quiz list, then createQuiz() to create a new quiz, updateQuiz() to update an existing quiz, and score() to monitor student performance before saving the quiz changes through the function saveQuiz(). Meanwhile, the student will select a quiz title and invoke the function attemptQuiz() to participate in the quiz, which will then trigger the function evaluateQuiz() to evaluate the quiz result, providing a badge for the student if the achievement criteria are met.

Sequence Diagram 7: Manage Class Management

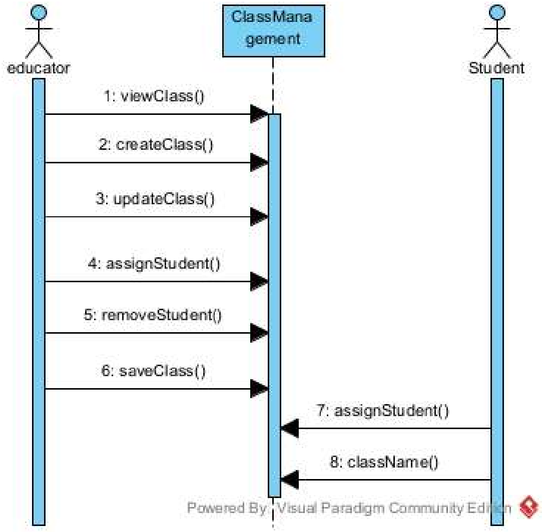


Figure 4.16: Sequence Diagram for Manage Class Management

Description: The diagram depicts a process flow on how classes are organized. The Educator initiates by checking the classes through viewClass(). The Educator can add a class through createClass() or edit a class through updateClass(). While updating a class, the Educator can manage the class by adding members through assignStudent() or deleting members through removeStudent(). The Educator then updates the class through saveClass(). The Student will be able to join a class through assignStudent() and can view the class through className().

4.4.6 CRUD Matrix

The CRUD Matrix shows how various actors are able to create, read, update, or delete the primary system entities present in NumSolve 2.0. The significance of this matrix is that all roles, whether students, teachers, or system analysts, are identified, making all functions related to secured access and workflows possible through this diagram alone.

Table 4.8: CRUD Matrix of NumSolve 2.0

Entity	Actions	C	R	U	D
User	Student and Educator register new accounts	x			
	Student and Educator view their profiles		x		
	Student and Educator update their profiles (and password)			x	
	System Analyst manages user accounts	x	x	x	x
Computation	Student and Educator execute new computation	x			
	Student and Educator view saved computations		x		
	Student and Educator export/save results			x	
	Student and Educator delete their computation records				x
Learn Material	Educator uploads new learning material	x			
	Student and Educator view learning materials		x		
	Educator updates existing learning materials			x	
	Educator deletes outdated learning materials				x
Quiz	Educator creates new quizzes	x			
	Student attempts quizzes (Read Questions)		x		
	Educator views quiz list and student scores		x		
	Educator updates quiz details			x	
	Educator deletes quizzes				x
Class Mgmt	Educator creates new classes	x			
	Student and Educator view class details		x		
	Educator updates class (assign/remove student)			x	
	Student joins class (Updates student list)			x	
	Educator deletes a class				x
Report	Student and Educator generate computation reports	x			
	Student, Educator, and Analyst view generated reports		x		
	System Analyst updates system-wide reports			x	
	System Analyst deletes reports				x
Note: C=Create, R=Read, U=Update, D=Delete					

CHAPTER 5

SYSTEM DESIGN

This Chapter presents how to transition from Functional Specification to Technical Specification for the NumSolve 2.0 Project. The document defines the high-level architecture (overview), logical organization (design) of software components, and database structure. This set of technical architectures helps provide the framework for later Implementation and ensures that the Implementation has a solid and scalable design.

5.1 Introduction

The phase of system design represents a major element of the software development life cycle by serving as a connecting link between the "what" of system requirements and the "how" of the technology to fulfill those requirements. Within the design development of NumSolve 2.0, the design process will include developing and creating the interdependent technical modules necessary to meet the functional needs (for example, intelligent method recommendations and iterative computation) of the system.

The main purpose of this phase is to create a model of the system's internal structure, including data flow and end-user interaction patterns, the ability to process complex mathematical algorithms accurately (with precise results) and the ability to create a web-based presence without excessive use of system resources. The following outlines the primary design methodologies used to create the system:

use of a three-tier architectural framework and the use of modular programming concepts in the creation of the system's technical modules. Ultimately, these design decisions have been made in order to provide end users (students and researchers) with a seamless, intuitive experience when utilising the system's capabilities to learn about numerical concepts, and free from technical complications. Having documented the design decisions allows the system to have a level of transparency and maintainability necessary to fulfil its long-term commitment to academic use.

5.2 System Architecture

NumSolve 2.0's architectural design gives you a basic overview of the design of different elements and how they work together to provide functional requirements. The architecture provides a more structured framework for developing a system to support complex calculations using mathematics and provide an easy-to-use interface. This section will provide the background for the decision, how it was organized into a tiered model, and why the decision was made in order to achieve long-term scalability and ongoing maintenance for the project.

5.2.1 Architectural Design

The architecture of the NumSolve 2.0 system is built using three-tier modeling for modularity and maintainability. The system is made up of three major components or subsystems, namely the Presentation Layer, Application Layer, and Database Layer, each carrying out its own functions in support of the overall system functionality.

The Presentation Layer concerns the representation of how users will interact with the app through the web browser. It is developed in a way to present screens for logging in, entering equations/datasets, choosing numerical methods, visualizing results graphically, and maintaining histories of computations, using JSP, HTML, CSS, and client-side scripts. This layer is crafted to be responsive and usable on both desktops and mobile devices.

The Application Layer is the central processing engine of the entire system. It deals with mathematical processing, recommendation of rule-based methods, user input validation, and control of all system flows. This particular level comprises the business logic of all mathematical methods, namely root detection, interpolation, differentiation, integration, and solution of ODE. It also deals with all interactions between the user interface and the database.

The Database Layer handles the storage of user information, computational records, learning resources, and produced reports. This layer makes use of the relational database management system and ensures the security and integrity of the data.

These subsystems work together in a controlled way, where the inputs provided by the Presentation Layer are handled by the Application Layer, which can retrieve or save data based on the requirement via the Database Layer. Such a way of layering the subsystem allows them to be developed and tested independently, although they act as a unit.

5.3 Design Rationale

The three-layer architecture was incorporated into the development of the NumSolve 2.0 because the system needed to incorporate the principles of modularity, usability, and scalability. Modular systems comprise three layers, including the user interface layer, the logic layer, and the data management layer.

This will help in achieving flexibility in terms of allowing changes in the user interface, such as enhanced visualization and/or layout improvements, without necessarily impacting the logic of numerical computations. At the same time, improvements in numerical algorithms and decision-making logic can be achieved without necessarily influencing storage and user interface aspects within the Application Layer. This would be critical for the proposed eduational platform considering its anticipated future enhancements in terms of additional numerical algorithms.

Other architectures explored included single-tier (also known as monolithic architecture) designs; these designs lack scalability as well as ease of maintenance. Two-tier architecture was also explored; in this kind of architecture, the business logic is tied to the user interface layer.

The chosen architecture also fits well with the educational goals of the system. The architecture promotes systematic learning process management, secure data management, and effective computation management, in addition to a clear division of tasks. On the whole, the three-tier architecture offers a fair mix that satisfies the current needs of the proposed project, along with flexibility for expansion.

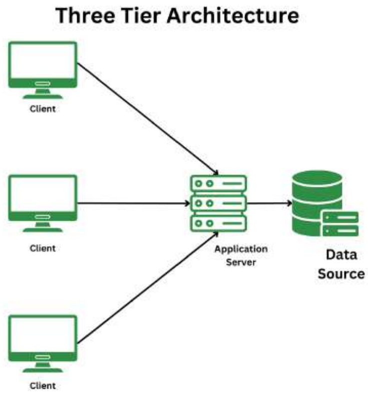


Figure 5.1: Three-Tier Architecture of NumSolve 2.0

5.4 Package Diagram

A Package Diagram is a structural diagram in the Unified Modeling Language (UML) that depicts the organization of a system into logical groupings or modules, known as packages. It illustrates the dependencies and relationships between these high-level system components, providing a macro-level view of the software architecture. By grouping related classes and entities, the package diagram helps manage system complexity, promotes modularity, and serves as a blueprint for development and deployment.

For the **NumSolve 2.0** system, the architecture is divided into six primary packages: User Management, Class Management, Learning Content, Communication, Assessment, and the Computation Engine. These modules are interconnected, reflecting the data flow and functional dependencies required to deliver a comprehensive educational and computational platform.

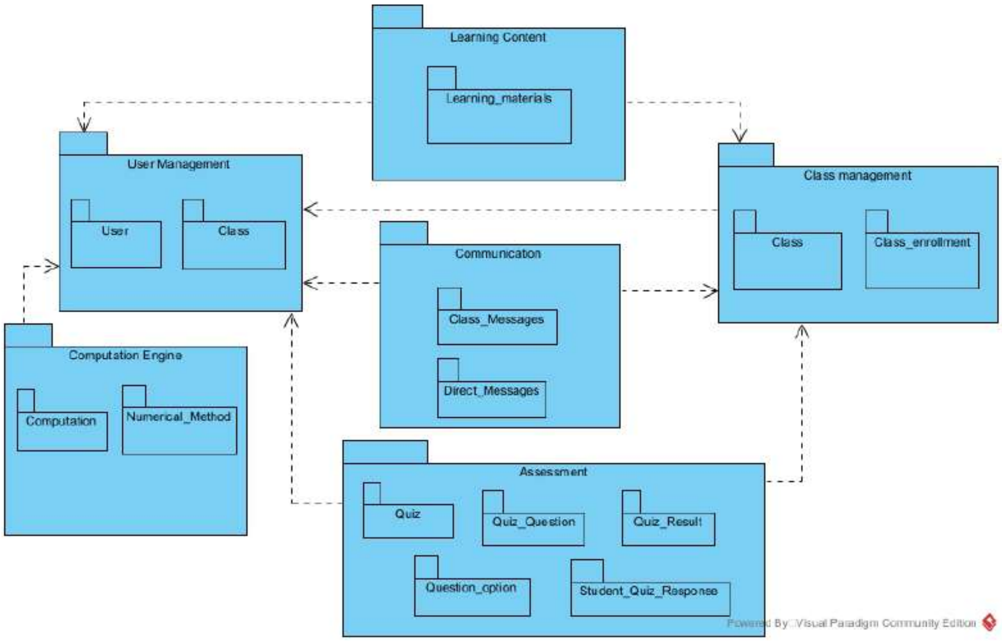


Figure 5.2: Package Diagram of NumSolve 2.0

5.4.1 Package Descriptions

The NumSolve 2.0 system is modularized into the following cohesive packages to ensure separation of concerns and maintainability:

- **User Management:** This package is the foundational module of the system, responsible for handling user authentication, profiles, and access control. It encapsulates entities related to users and their respective roles (e.g., Student, Educator). Most other packages in the system depend on User Management to identify who is performing an action.

- **Class Management:** This package handles the creation, organization, and tracking of educational cohorts. It includes the *Class* and *Class_enrollment* components. It relies on User Management to associate classes with educators and enrollments with students.

- **Learning Content:** Dedicated to the distribution of educational resources, this package manages the *Learning_materials*. It is dependent on both User

Management (to track the uploader) and Class Management (to link materials to specific classes).

- **Communication:** This module facilitates interactions between users within the platform. It contains components for *Class_ Messages* and *Direct_ Messages*. It depends heavily on Class Management for group contexts and User Management for sender/receiver identification.

- **Assessment:** A comprehensive module designed to handle system evaluations. It encapsulates the entire quiz lifecycle, including *Quiz*, *Quiz_ Question*, *Question_ option*, *Student_ Quiz_ Response*, and *Quiz_ Result*. It interfaces with Class Management to assign quizzes to specific classes and User Management to track student scores.

- **Computation Engine:** Serving as the core technical feature of NumSolve 2.0, this package executes mathematical processing. It groups the *Computation* histories and the library of *Numerical_ Method* algorithms. It depends on User Management to log computation histories to specific users.

5.5 Database Design

5.5.1 Normalization

Normalization is the process of structuring data within the database in an effort to eliminate redundancy and improve data integrity. Normalization ensures data dependency is sound so that modification, insertion, or deletion is done without any anomalies. At the start of this project, the database design for the NumSolve 2.0 system has been normalized up to Third Normal Form (3NF) for consistency and scalability.

Zero Normal Form: 0NF

In Zero Normal Form (0NF), data is stored in an unstructured way where it is possible to have repeating groups and multi-valued attributes. The entire system has all information stored in only one table, leading to redundancy and difficulties in maintenance of databases.

For NumSolve 2.0, the unnormalized form involves the combination of user information, computation methods, computations, reports, and learning resources in the same table. Here, there is the repetition of user information, names of methods, and learning resources in each computation.

0NF

User_id	**FullName**	**UserName**	**Password**	**UserRole**	**Email**
U01	Ali Bin Abu	Ali	**********	Student	ali@gmail.com
Phone	**Location**	**Department**	**Bio**	**MemberSince**	**MethodName**
010-3399396	Kuala Lumpur	IOT	keep smilimng	2025-1-23	Bisection, secant
InputData	**Result**	**ErrorValue**	**Iteration**	**GraphType**	**ReportFormat**
x^2-4	2	0.001	5	line	PDF
QuizTitle	Score	className	ClassCode	message_body	sent_at
Bisection	2	K1	1234	best tak	22 Mac
option_text	is_correct	question_text	explanation	points	attempt_date
what what	yes	apa itu makanan	makanan ialah food	10	2025-01-10

Figure 5.3: 0NF for NumSolve 2.0

5.5.2 One Normal Form: 1NF

First Normal Form (1NF) specifies that all the attributes in the table should have atomic values. Every record in the table should be able to be uniquely identified. For the data to be in 1NF, it is organized again in such a way that all the groups and repetitive data are eliminated. For the data to be unique, it is assigned an identifier, computation ID. Redundancy still occurs since data is stored twice in the form of user details and mathematical method descriptions.

1NF

UserID	**FullName**	**UserName**	**Password**	**UserRole**	**Email**
U01	Ali Bin Abu	Ali	**********	Student	ali@gmail.com
Phone	**Location**	**Department**	**Bio**	**MemberSince**	**MethodName**
010-3399396	Kuala Lumpur	IOT	keep smilimng	2025-1-23	Bisection
InputData	**Result**	**ErrorValue**	**Iteration**	**GraphType**	**ReportFormat**
x^2-4	2	0.001	5	line	PDF
QuizTitle	Score	className	ClassCode	message_body	sent_at
Bisection	2	K1	1234	best tak	22 Mac
option_text	is_correct	question_text	explanation	points	attempt_date
what what	yes	apa itu makanan	makanan ialah food	10	2025-01-10

Figure 5.4: 1NF for NumSolve 2.0

5.5.3 Second Normal Form: 2NF

Second Normal Form (2NF): This form improves 1NF by removing partial dependency. At this point, all non-key attributes of a table must fully depend on the primary key. One way to achieve this is by dividing the database into different tables for users, computations, and numerical methods. The details of users are maintained in a separate table from computation information, while information about numerical methods should be maintained in a separate table. In this form, every attribute completely depends on its respective primary key.

2NF

User_ID	Full_Name	UserName	UserRole	Email	Phone
U01	Ali Bin Abu	Ali	Student	ali@gmail.com	010-3399396

Location	Department	Bio	MemberSince	PhotoPath
Kuala Lumpur	IOT	keep smilimng	2025-1-23	file/1.png

Figure 5.5: 2NF table for user in NumSolve 2.0

Method_ID	Method_Name	Category	Description
M01	Bisection	Root Finding	Narrow interval
M02	Secant	root Finding	secant line to root

Figure 5.6: 2NF table for method in NumSolve 2.0

Computation_ID	User_ID	Method_ID	Input_Data	Result
C01	U01	M01	x^2-4	2

ErrorValue	Iteration	GraphType	ReportFormat	Title	Description
0.001	5	line	PDF	numerical	math things

Figure 5.7: 2NF table for computation in NumSolve 2.0

class_id	class_name	class_code	class_description	user_id	created_date
CC1	K1	1234	bisection class	U01	2025-01-10

Figure 5.8: 2NF table for Class in NumSolve 2.0

enrollment_id	class_id	user_id	enroll_date
E01	CC1	U01	2025-01-10

Figure 5.9: 2NF table for Enrollment_class in NumSolve 2.0

group_message_id	class_id	sender_id	message_body	sent_at
GM1	CC1	U01	best tak	22 MAC

Figure 5.10: 2NF table for class_message in NumSolve 2.0

message_id	class_id	sender_id	receiver_id	message_body	sent_at
ME1	CC1	U01	U02	best tak	22 Mac

Figure 5.11: 2NF table for Direct_messages in NumSolve 2.0

quiz_id	quiz_title	quiz_description	visibility	time_limit	total_marks
Q01	Bisection	know more	private	30	10

created_date	user_id	class_id	PhotoPath
2025-01-10	U01	CC1	files/int.MP4

Figure 5.12: 2NF table for Quiz in NumSolve 2.0

response_id	result_id	question_id	selection_option_id	quiz_id	User_id
R1	RR01	Q1	SO1	Q01	U01

total_marks	attempt_date	question_id	selected_option_id	is_correct
10	2025-01-10	Q1	SO1	yes

Figure 5.13: 2NF table for Student_response in NumSolve 2.0

question_id	quiz_id	question_text	explanation	points
Q1	Q01	apa itu makanan	makanan ialah food	10

Figure 5.14: 2NF table for Quiz_question in NumSolve 2.0

selected_option_id	question_id	option_text	is_correct
SO1	Q1	what what	yes

Figure 5.15: 2NF table for question_option in NumSolve 2.0

Material_id	User_id	Topic	Description_material	Material_Type	Upload_Date	FilePath	photoPath_materials	file_Name	File_size	Class_id
LM01	U02	Integration	Video for student	Video	2025-01-10	files/int.MP4	file/ image.png	method	231	CC1

Figure 5.16: 2NF table for Material in NumSolve 2.0

5.5.4 Third Normal Form: 3NF

Third Normal Form, denoted as 3NF, removes transitive dependency. The non-key columns are not dependent on other non-key columns. To achieve this in NumSolve 2.0, there is a further breakup of the related concepts into different entities such as roles, reports, numerical methods, and learning materials. The tables are designed in such a manner that the columns are dependent only on the primary key. The resulting design is free from redundancy, scalable, and robust for system operations as well as for further system improvement.

User_id	Full_Name	UserName	Role_id	Email	Phone	Location	Department	Bio	MemberSince	PhotoPath	password	last_login_date
U01	Ali Bin Abu	Ali	R01	ali@gmail.com	010-3399396	Kuala Lumpur	IOT	keep smilimng	2025-1-23	file/1.png	ali123	31-05-2026

Figure 5.17: 3NF table for user in NumSolve 2.0

Role_id	Role_Name	Description_role
R01	Student	learn methods
R02	Educator	teach student
R03	Administrator	manage system

Figure 5.18: 3NF table for Role in NumSolve 2.0

Method_id	Method_Name	Category	Description_method
M01	Bisection	Root Finding	Narrow interval
M02	Secant	root Finding	secant line to root

Figure 5.19: 3NF table for numerical_Method in NumSolve 2.0

Computation_id	User_id	Method_id	Input_data	Result
C01	U01	M01	x^2-4	2

ErrorValue	Iteration	GraphType	Computation_date	Title	Description
0.001	5	line	PDF	numerical	math things

Figure 5.20: 3NF table for Computation in NumSolve 2.0

Material_id	User_id	Topic	Description_material	Material_Type	Upload_Date	FilePath	photoPath_materials
LM01	U02	Integration	Video for student	Video	2025-01-10	files/int.MP4	file/ image.png

file_Name	File_size	Class_id
method	231	CC1

Figure 5.21: 3NF table for Material in NumSolve 2.0

class_id	class_name	class_code	class_description	user_id	created_date
CC1	K1	1234	bisection class	U01	2025-01-10

Figure 5.22: 3NF table for class in NumSolve 2.0

enrollment_id	class_id	user_id	enroll_date
E01	CC1	U01	2025-01-10

Figure 5.23: 3NF table for class_Enrollment in NumSolve 2.0

quiz_id	quiz_title	quiz_description	visibility	time_limit	total_marks
Q01	Bisection	know more	private	30	10

created_date	user_id	class_id	PhotoPath
2025-01-10	U01	CC1	files/int.MP4

Figure 5.24: 3NF table for Quiz in NumSolve 2.0

result_id	quiz_id	user_id	score	attempt_date
RR01	Q01	U01	2	2025-01-10

Figure 5.25: 3NF table for quiz_result in NumSolve 2.0

question_id	quiz_id	question_text	explanation	points
Q1	Q01	apa itu makanan	makanan ialah food	10

Figure 5.26: 3NF table for quiz_question in NumSolve 2.0

group_message_id	class_id	sender_id	message_body	sent_at
GM1	CC1	U01	best tak	22 MAC

Figure 5.27: 3NF table for class_messages in NumSolve 2.0

response_id	result_id	question_id	selection_option_id
R1	RR01	Q1	SO1

Figure 5.28: 3NF table for student_quiz_response in NumSolve 2.0

selected_option_id	question_id	option_text	is_correct
SO1	Q1	what what	yes

Figure 5.29: 3NF table for direct_message in NumSolve 2.0

message_id	class_id	sender_id	receiver_id	message_body	sent_at
ME1	CC1	U01	U02	best tak	22 Mac

Figure 5.30: 3NF table for user in NumSolve 2.0

5.6 Entity Relational Diagram

An Entity Relationship Diagram (ERD) is a type of diagram that lets you see how different entities (e.g. people, customers, or other objects) relate to each other in an application or a database.

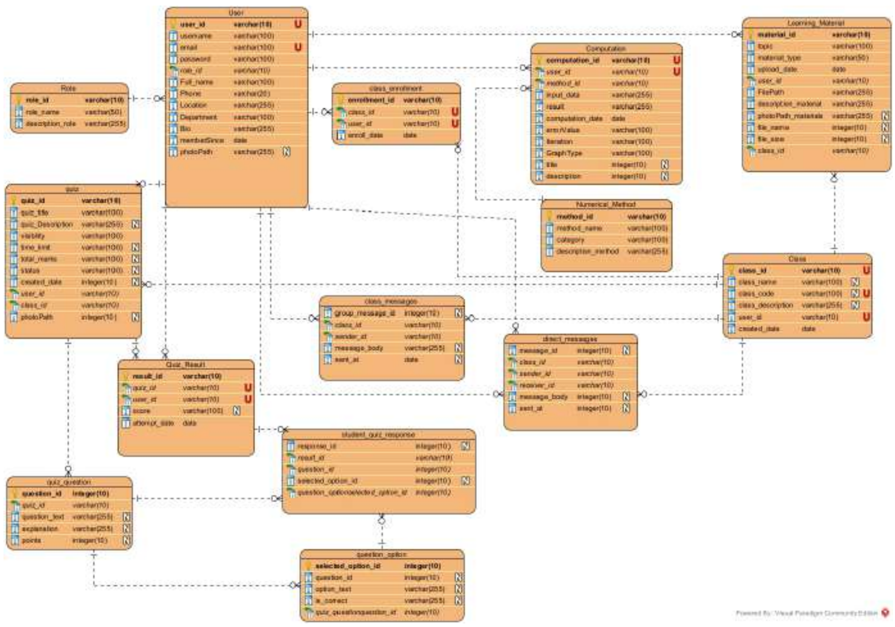


Figure 5.31: Entity Relational Diagram of NumSolve 2.0

5.7 Data Dictionary

The following data dictionary encompasses all key system entities for the **NumSolve 2.0** system in an alphabetical order format to ensure clarity and ease of reference. Entities within this data dictionary are described in terms of their structure, which encompasses various characteristics, as well as a function within the system, to create an appropriate reference for database implementation and future system improvement, in conformity with previous database design and normalization outputs.

5.7.1 Class

The **Class** entity acts as the primary structural container for educational environments within the platform. It allows educators to create designated virtual rooms where learning materials, quizzes, and group discussions are localized. Each record holds the course's descriptive metadata and unique registration code required for student access.

Table 5.1: Class Entity

Attribute Name	**Data Type**	**Description**
class_id	VARCHAR(10) (PK)	Unique identifier for a class
class_name	VARCHAR(100)	Name of the class
class_code	VARCHAR(100)	Registration code for the class
class_description	VARCHAR(255)	Description of the class
user_id	VARCHAR(10) (FK)	References the educator who created the class
created_date	DATE	Date the class was created

5.7.2 Class_Enrollment

The **Class_Enrollment** entity is an associative (junction) table essential for resolving the many-to-many relationship between users and classes. It tracks which students are members of which classes, serving as the basis for access control regarding class-specific materials, quizzes, and messaging.

Table 5.2: Class_Enrollment Entity

Attribute Name	**Data Type**	**Description**
enrollment_id	VARCHAR(10) (PK)	Unique identifier for the enrollment
class_id	VARCHAR(10) (FK)	References the enrolled class
user_id	VARCHAR(10) (FK)	References the enrolled student
enroll_date	DATE	Date of enrollment

5.7.3 Class_Messages

The **Class_Messages** entity facilitates collaborative group communication. It stores the chronological history of broadcast messages sent within a specific class environment, linking the text content to the sender and the designated classroom to ensure messages remain contextually isolated.

Table 5.3: Class_Messages Entity

Attribute Name	**Data Type**	**Description**
group_message_id	INTEGER(10) (PK)	Unique identifier for a group message
class_id	VARCHAR(10) (FK)	References the class the message belongs to
sender_id	VARCHAR(10) (FK)	References the user who sent the message
message_body	VARCHAR(255)	The text content of the message
sent_at	DATE	Date and time the message was sent

5.7.4 Computation

The **Computation** entity represents the technical core of the NumSolve 2.0 system. It functions as a historical execution log, recording every mathematical calculation performed by a user. It captures the initial input variables, the specific numerical algorithm utilized, iteration constraints, and the final computed result.

Table 5.4: Computation Entity

Attribute Name	**Data Type**	**Description**
computation_id	VARCHAR(10) (PK)	Unique identifier for each computation record
user_id	VARCHAR(10) (FK)	References the user who performed the computation
method_id	VARCHAR(10) (FK)	References the numerical method used
input_data	VARCHAR(255)	Equation or dataset provided by the user
result	VARCHAR(255)	Computed numerical result
computation_date	DATE	Date when computation was executed
errorValue	VARCHAR(100)	Error or tolerance value of the computation
Iteration	VARCHAR(100)	Number of iterations performed
GraphType	VARCHAR(100)	The type of graph used
title	VARCHAR(100)	Computation title reference
description	VARCHAR(255)	Computation description reference

5.7.5 Direct_Messages

The **Direct_Messages** entity governs secure, one-on-one communication between distinct users. Unlike group discussions, this table strictly links a single sender to a single receiver, allowing for private inquiries, feedback, or peer-to-peer assistance within the platform.

Table 5.5: Direct_Messages Entity

Attribute Name	**Data Type**	**Description**
message_id	INTEGER(10) (PK)	Unique identifier for a direct message
class_id	VARCHAR(10) (FK)	Contextual class reference for the message
sender_id	VARCHAR(10) (FK)	References the user who sent the message
receiver_id	VARCHAR(10) (FK)	References the user receiving the message
message_body	VARCHAR(255)	Content of the message
sent_at	DATE	Timestamp of the message

5.7.6 Learning_Material

The **Learning_Material** entity is responsible for managing the educational resources distributed by instructors. It acts as a digital library index, storing essential file paths, metadata, format types, and descriptions for documents or media uploaded to a specific class.

Table 5.6: Learning_Material Entity

Attribute Name	**Data Type**	**Description**
material_id	VARCHAR(10) (PK)	Unique identifier for learning material
topic	VARCHAR(100)	Title of the learning material
material_type	VARCHAR(50)	Type of the learning content
upload_date	DATE	Date when the material was uploaded
user_id	VARCHAR(10) (FK)	Educator who uploaded the material
class_id	VARCHAR(10) (FK)	Associated class for the material
FilePath	VARCHAR(255)	Location of the uploaded file
description_material	VARCHAR(255)	Description of the materials
photoPath_materials	VARCHAR(255)	Location of photo uploaded
file_name	VARCHAR(255)	Reference for the file name
file_size	VARCHAR(50)	Size of the uploaded file

5.7.7 Numerical_Method

The **Numerical_Method** entity operates as a static reference catalog within the system. It defines the library of available mathematical algorithms (such as root finding or numerical integration techniques) that users can select when initiating a computation.

Table 5.7: Numerical_Method Entity

Attribute Name	**Data Type**	**Description**
method_id	VARCHAR(10) (PK)	Unique identifier for numerical method
method_name	VARCHAR(100)	Name of the numerical method
category	VARCHAR(100)	Category of the method
description_method	VARCHAR(255)	Explanation of the method

5.7.8 Question_Option

The **Question_Option** entity defines the multiple-choice structure for assessments. For every question dynamically generated in a quiz, this table holds the possible answer choices. It securely isolates the "correct" flag to allow automated system grading.

Table 5.8: Question_Option Entity

Attribute Name	**Data Type**	**Description**
selected_option_id	INTEGER(10) (PK)	Unique identifier for a question option
question_id	INTEGER(10) (FK)	References the parent quiz question
option_text	VARCHAR(255)	Text content of the option
is_correct	VARCHAR(255)	Indicator if the option is the correct answer

5.7.9 Quiz

The **Quiz** entity acts as the main wrapper for all system evaluations. It tracks the macro-level configuration of an assessment, such as the title, active status, cumulative maximum marks, and predefined time limits, linking the evaluation directly to an ongoing class.

Table 5.9: Quiz Entity

Attribute Name	Data Type	Description
quiz_id	VARCHAR(10) (PK)	Unique identifier for a quiz
quiz_title	VARCHAR(100)	Title of the quiz
quiz_Description	VARCHAR(255)	Detailed description of the quiz
visibility	VARCHAR(100)	Visibility status of the quiz
time_limit	VARCHAR(100)	Time limit to complete the quiz
total_marks	VARCHAR(100)	Maximum score possible
status	VARCHAR(100)	Active or inactive status
created_date	DATE	Date the quiz was created
user_id	VARCHAR(10) (FK)	References the user who created it
class_id	VARCHAR(10) (FK)	References the associated class
photoPath	VARCHAR(255)	Location of a cover image for the quiz

5.7.10 Quiz_Question

The **Quiz_Question** entity holds the individual queries that make up a full assessment. Connected dependently to the Quiz table, it stores the exact textual prompt, the designated point value for a correct response, and a descriptive explanation that can be shown to students post-evaluation.

Table 5.10: Quiz_Question Entity

Attribute Name	Data Type	Description
question_id	INTEGER(10) (PK)	Unique identifier for a question
quiz_id	VARCHAR(10) (FK)	References the parent quiz
question_text	VARCHAR(255)	Content of the question
explanation	VARCHAR(255)	Explanation of the correct answer
points	INTEGER(10)	Point value for the question

5.7.11 Quiz_Result

The **Quiz_Result** entity captures the final outcome of a student's assessment attempt. As a top-level transactional record, it securely stores the total aggregated score achieved by the user on a specific date, allowing educators to track overall academic performance.

Table 5.11: Quiz_Result Entity

Attribute Name	Data Type	Description
result_id	VARCHAR(10) (PK)	Unique identifier for a quiz attempt result
quiz_id	VARCHAR(10) (FK)	References the attempted quiz
user_id	VARCHAR(10) (FK)	References the student taking the quiz
score	VARCHAR(100)	Achieved score
attempt_date	DATE	Date the quiz was taken

5.7.12 Role

The **Role** entity is a critical component of the platform's security and authorization architecture. By abstracting user privileges (e.g., Student vs. Educator) into a distinct table, it enforces robust Role-Based Access Control (RBAC) across the entire application ecosystem.

Table 5.12: Role Entity

Attribute Name	Data Type	Description
role_id	VARCHAR(10) (PK)	Unique identifier for user role
role_name	VARCHAR(50)	Role name (Student, Educator, Analyst)
description_role	VARCHAR(255)	Description of each role

5.7.13 Student_Quiz_Response

The **Student_Quiz_Response** entity provides highly granular tracking of examination data. It maps a specific user's quiz attempt directly to the exact multiple-choice option they selected for each question, allowing for deep statistical analysis on where students commonly struggle.

Table 5.13: Student_Quiz_Response Entity

Attribute Name	Data Type	Description
response_id	INTEGER(10) (PK)	Unique identifier for the answer response
result_id	VARCHAR(10) (FK)	References the parent quiz result
question_id	INTEGER(10) (FK)	References the question being answered
selected_option_id	INTEGER(10) (FK)	References the option chosen by the student

5.7.14 User

The **User** entity represents the foundational identity layer of NumSolve 2.0. It acts as the central hub for all platform interactions, securely storing authentication credentials, personal demographic information, and visual profile data required to personalize the learning experience.

Table 5.14: User Entity

Attribute Name	Data Type	Description
user_id	VARCHAR(10) (PK)	Unique identifier for system user
username	VARCHAR(100)	Profile username
email	VARCHAR(100)	User email address
password	VARCHAR(100)	Encrypted user password
role_id	VARCHAR(10) (FK)	References user role
Full_name	VARCHAR(100)	Full name of the user
Phone	VARCHAR(20)	Phone number to contact
Location	VARCHAR(255)	Address of user
Department	VARCHAR(100)	Department of user
Bio	VARCHAR(255)	A brief introduction of the user
memberSince	DATE	The date they joined the system
photoPath	VARCHAR(255)	Location of profile photo

5.8 Interface Design

5.8.1 Overview of User Interface

The user interface of NumSolve 2.0 has been designed in a manner that it is intuitive, interactive, and user-friendly so that the user can efficiently carry out their numerical analysis work with minimal efforts and learning. From a user perspective, this system offers a convenient workflow that helps guide them from data input to interpretation of the result.Upon the successful initialization of the system, the user is directed to the login or registration page. After the successful completion of the process, the user is able to access the main dashboard, which is the hub of the entire application. The dashboard gives the user rapid access to the entire critical functionality, which comprises numerical computation, method recommendation, visualization, record management, and report creation. The users are able to carry out calculations by entering equations and data in structured forms. The system can use various methods of numerics and offers assistance in the form of guidance for the entry of parameters like the initial interval, tolerance, and iteration limit. The process of calculating results is based on the data entered by the users, and the results are obtained in the form of graphical outputs to display convergence.

The system also offers feedback capabilities for users using error messages, validations, and status notifications. For example, the user gets notifications about whether the inputs provided are incorrect, whether the method has failed to converge, or whether a calculation has been completed. Additionally, educational feedback like step-by-step explanations and theoretical remarks is provided for improving the understanding of the user about the employed numerical technique.Users also have the ability to manage their computation result records using a records interface, which provides them with options of viewing, editing, deleting, and exporting their saved records. They can export their reports and graphs in PDF and

CSV formats, respectively, for documentation and further analysis. In general, the user interface focuses on providing clarity, responsiveness, and educational worth, which ensures users accomplish expected tasks with meaningful responses.

5.8.2 Screen Images

To show the assessment tracking and expanded features of NumSolve 2.0, the images below detail the system functions, ranging from authentication to the new gamification modules.

Landing Page

The landing page is the main entry point for the application. It gives access to the login mechanism and the general information about the system.

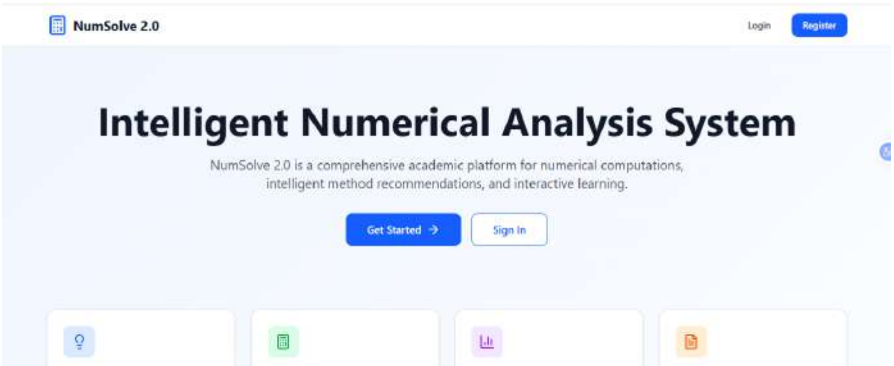


Figure 5.32: Landing Page

Login Screen

The login screen enables registered users to login in by entering their email address, and password. The system checks these entries against the existing records of users. A "Remember Me" option allows users to securely remain logged in. Successful Authentication redirects the user to the main dashboard, while invalid credentials trigger an error notification.

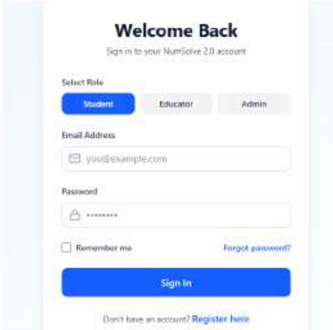


Figure 5.33: Login Screen Interface

Registration Screen

The registration interface enables new users to register. It is recorded that the system captures essential details like the complete name, email, and password. Most importantly, the user must choose their role (Student or Educator), which will dictate their access rights and dashboard layout.

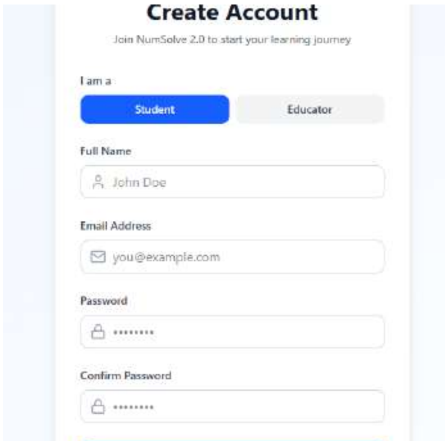


Figure 5.34: User Registration Interface

Student Dashboard

The Student Dashboard is the central place where all the activities of a student can be found. It provides immediate access to key functionalities like numerical analysis and method recommendation educational resources, and computation history. Widgets offer a brief summary of recent activity and statistics.

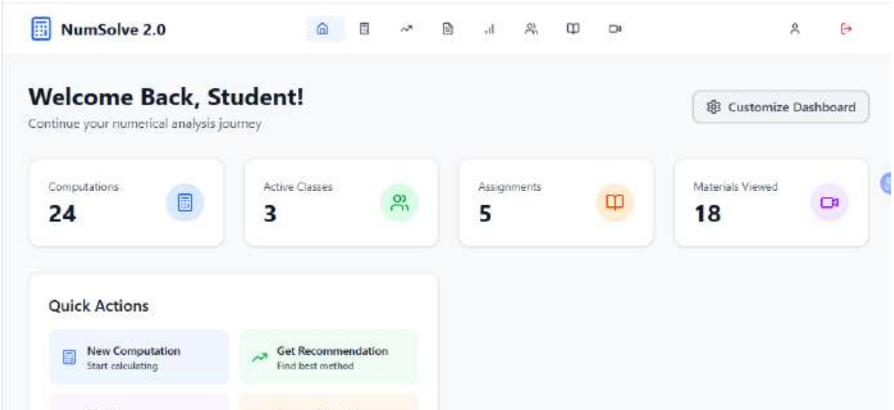


Figure 5.35: Student Dashboard Interface

Numerical Computation Interface

This interface enables users to carry out specific types of calculations. Users can specify a number of different calculations, including the numerical method, the algorithm to be used, and parameters such as tolerance levels, range, and iteration.

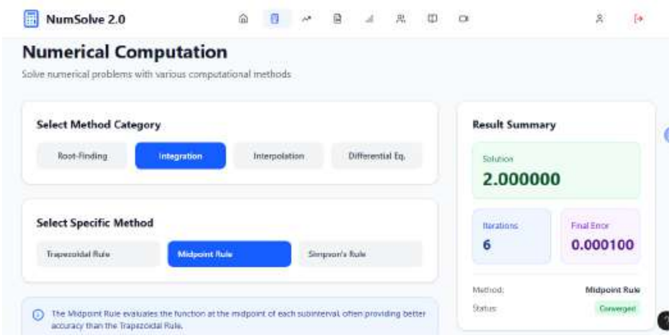


Figure 5.36: Numerical Computation Interface

Function Visualization Graph

The visualization screen represents the result of the numerical computation. It will show how functions behave, such as the convergence of results. The user interface is interactive, as users are allowed to zoom in, move horizontally, or highlight points for coordinate representations.

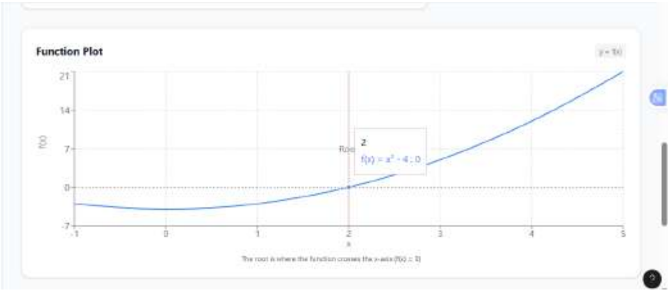


Figure 5.37: Function Visualization Graph

Method Recommendation Interface

This is a highly intelligent feature that assists the user in the selection of the "most appropriate" method of solution. It does this by asking the user for the equation, the type of problem, and the desired accuracy, which is then evaluated through the rule-based logic of the system.

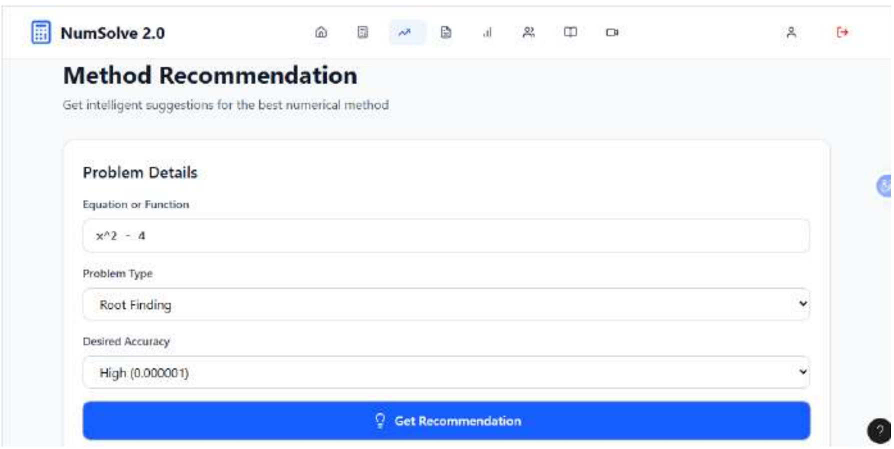


Figure 5.38: Method Recommendation Input

Method Recommendation Result

Based on the input analysis, the preferred approach, along with alternatives, is presented on this screen. It would enumerate the particular advantages of the preferred method so the user could directly exploit it on a new computation.

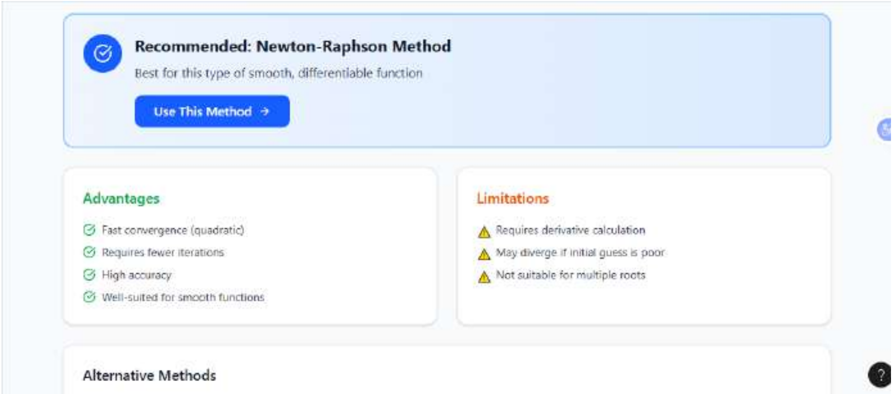


Figure 5.39: Method Recommendation Results

Computation Records

The records screen acts as a history log for users, enabling them to go through all numerical computation records stored in the system. The records screen provides users with the ability to search for, access, edit, or delete the stored records, enabling users to track their academic progress.

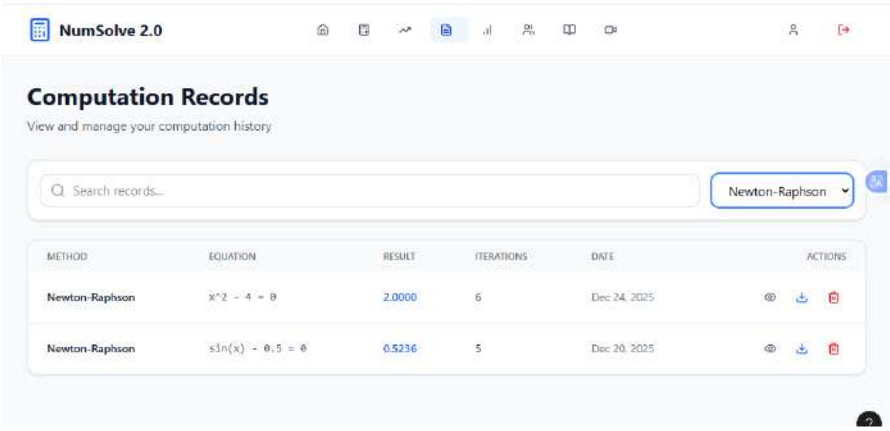


Figure 5.40: Computation Records

Report Generation Export

This interface allows a user to pick specific computation records, which can be used in creating a comprehensive report. A user has the option to personalize the specific data point he or she wants before exporting the final document intended for academic purposes.

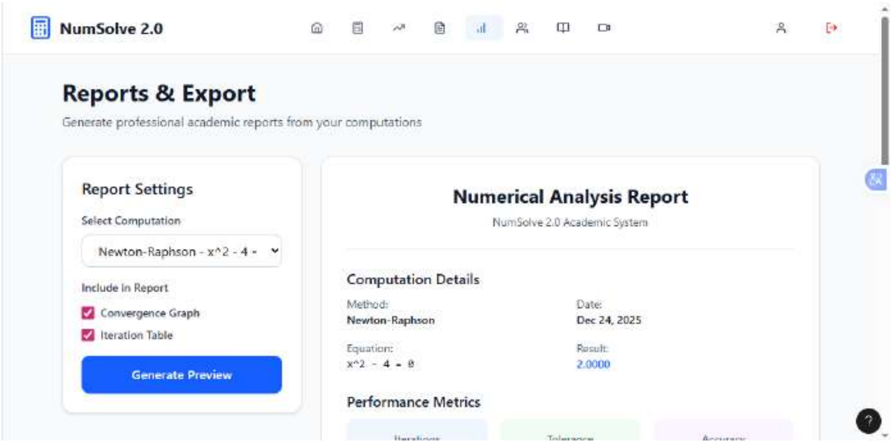


Figure 5.41: Report Generation Export

Computation Preview

Once the export is decided, the preview screen also provides the facility of viewing the report, which is done before any export is finalized. This ensures that all the figures, graphs, and explanations have been correctly placed. Later, the file can be downloaded in the form of PDF or CSV.

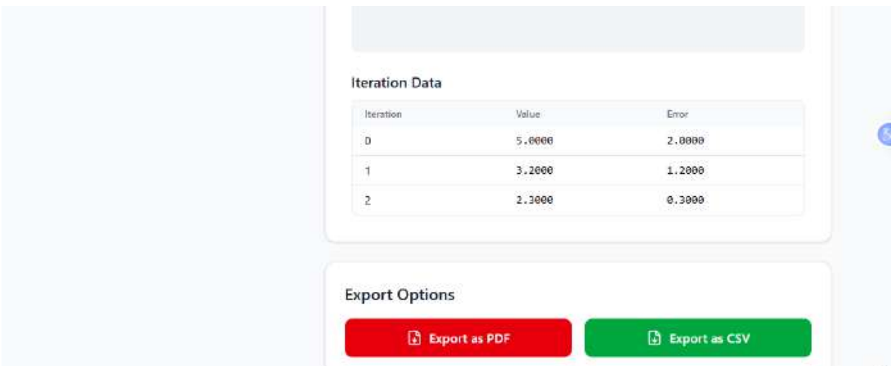


Figure 5.42: Report Preview

Learning Materials

The library contains educational material which can be accessed by the students. They can access lecture notes, examples, and video tutorials uploaded by the educators. There are various options for searching and filtering the material.

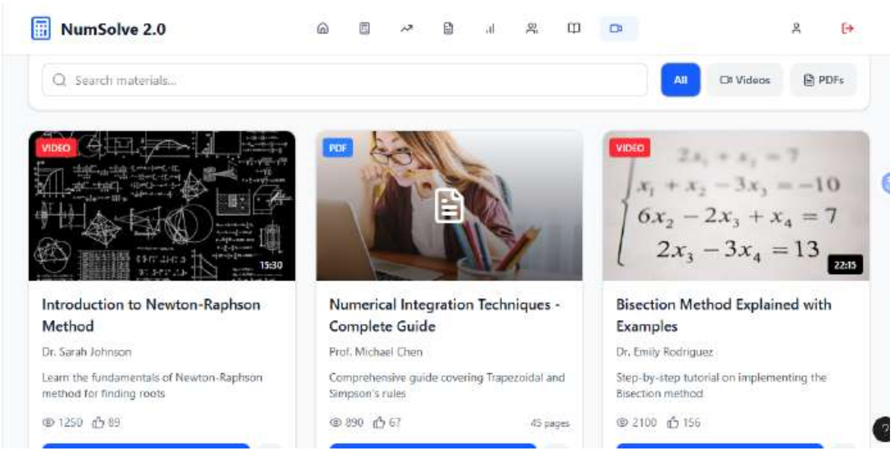


Figure 5.43: Learning Materials

My Classes (Student View)

The 'My Classes' interface enables students to search for and join specific courses using the class code. They can view all enrolled classes and navigate to specific courses as well as instructors.

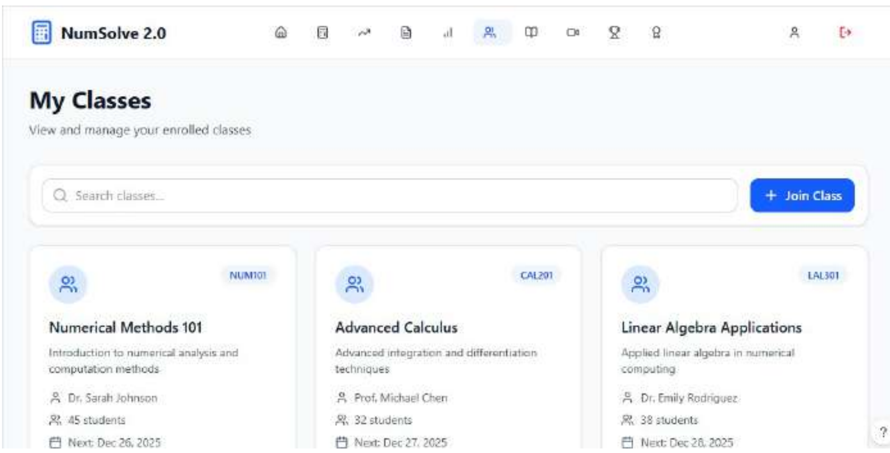


Figure 5.44: Student Class Management

Gamified Quizzes

In this section, the elements of gamification are integrated into the learning process. Students are given the option to view quizzes under different categories, track the points awarded to them, and view their current ranking.

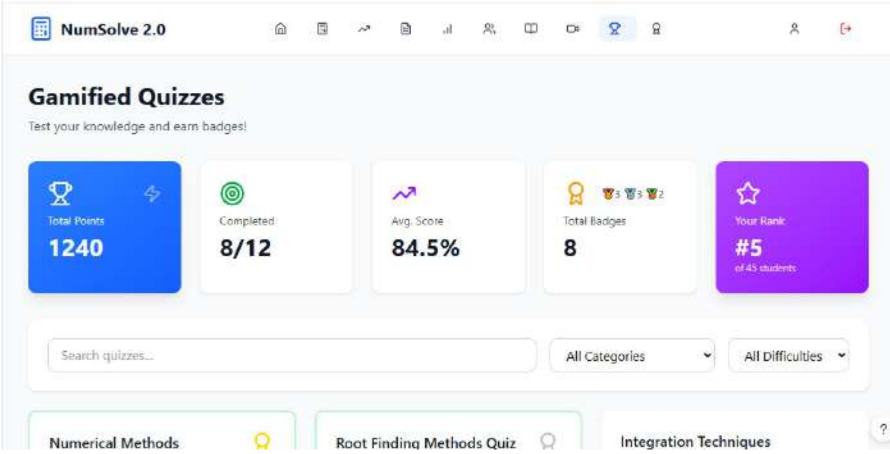


Figure 5.45: Gamified Quiz Dashboard

Leaderboard

The leaderboard encourages healthy competition by showing the top-performing students with regard to their quiz scores and engagement points. The top three are given visual prominence, and the rest follow after scrolling through the list.

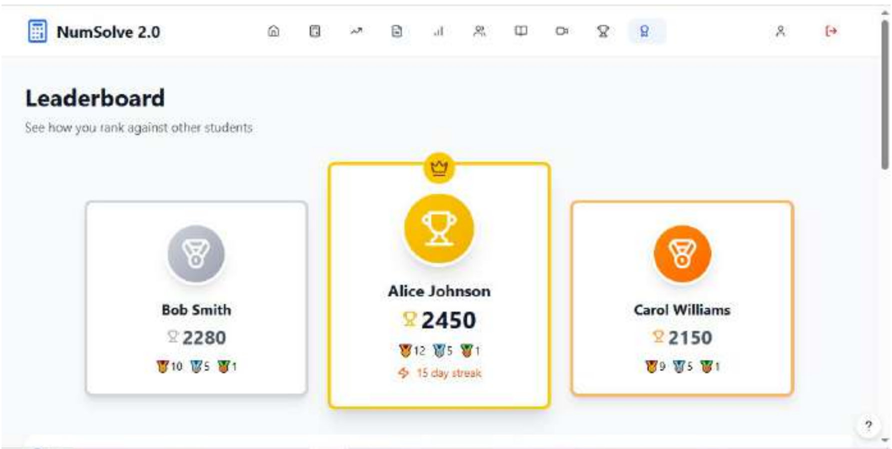


Figure 5.46: Student Leaderboard

User Profile

The profile screen will allow the user to manage his personal information, such as contact information and biography. The user will be able to update his data and save the changes; this securely keeps data in the system database.

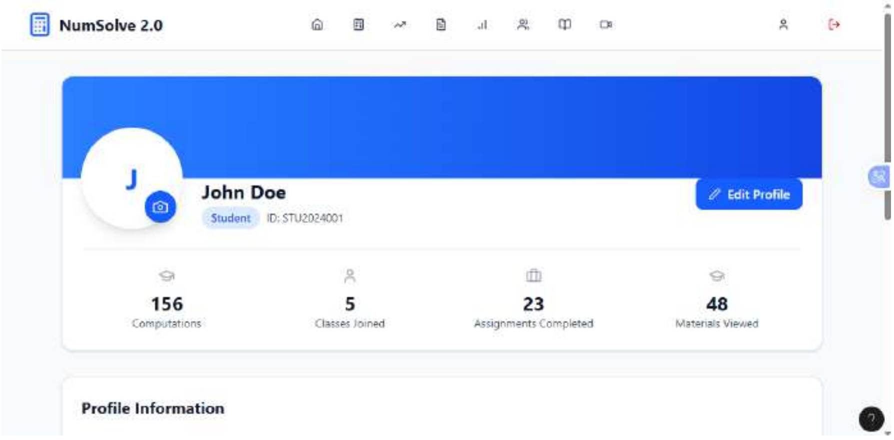


Figure 5.47: User Profile

Educator Dashboard

The Educator Dashboard is the teacher's hub. It allows teachers to create classes, learning materials, track students' activities, as well as view statistics on the activities of the courses on the system.

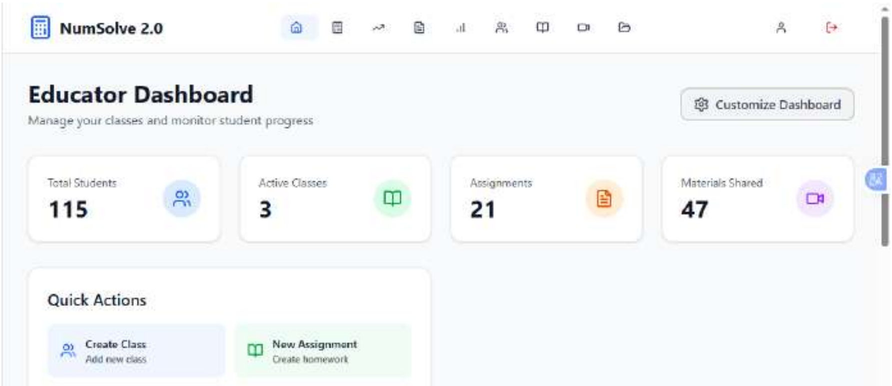


Figure 5.48: Educator Dashboard

Educator Class Management

This interface provides an educator with the ability to create a new class, edit existing ones, and monitor the number of enrolled students.

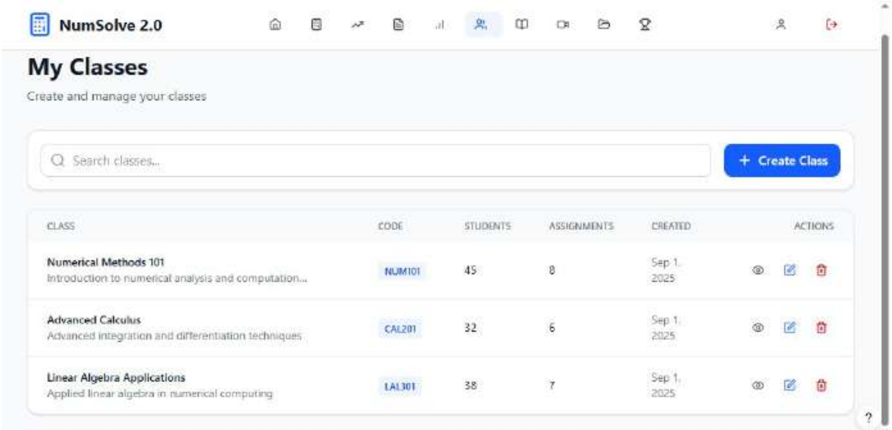


Figure 5.49: Educator Class Management

Quiz Management

Educators use this screen to create and manage gamified quizzes. The screen displays information about quizzes already published, student attempts, and the averages, thus helping to evaluate the level of comprehension of the students.

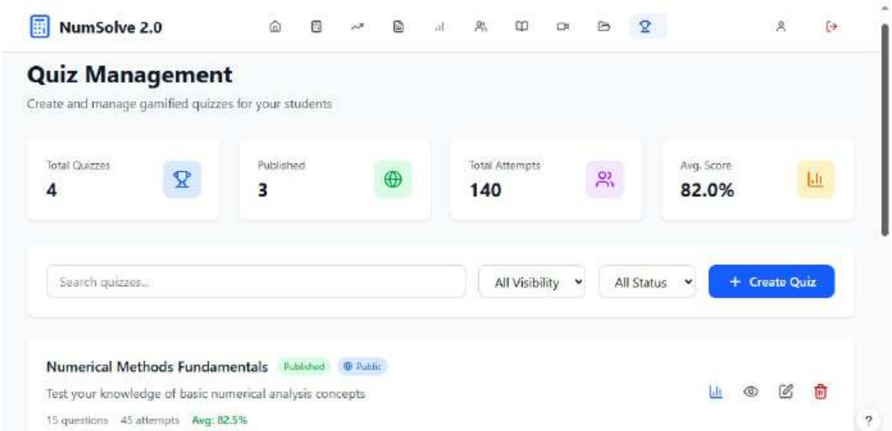


Figure 5.50: Educator Class Management

Educator Materials Management

This screen allows for the upload, modification, and deletion of learning resources. This enables educators to organize documents and media files, ensuring that learners access updated learning resources.

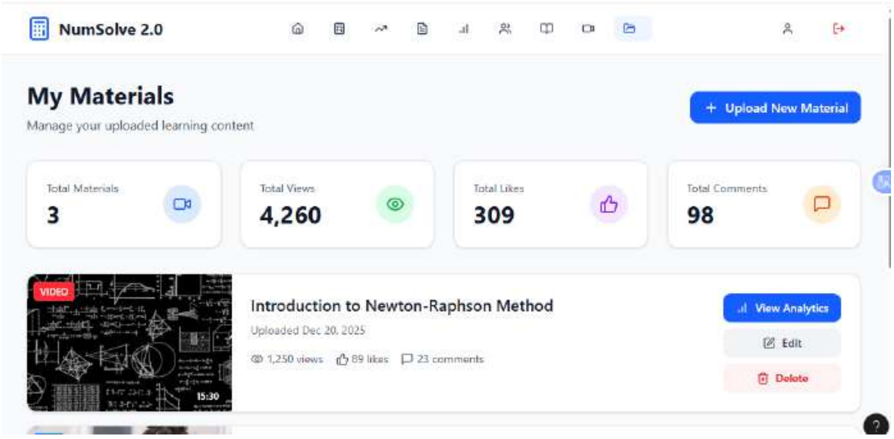


Figure 5.51: Educator Materials Management

System Administration Dashboard

The admin dashboard gives administrators an overall management view of the system. It can be used to manage user accounts, view log reports, and monitor system status.

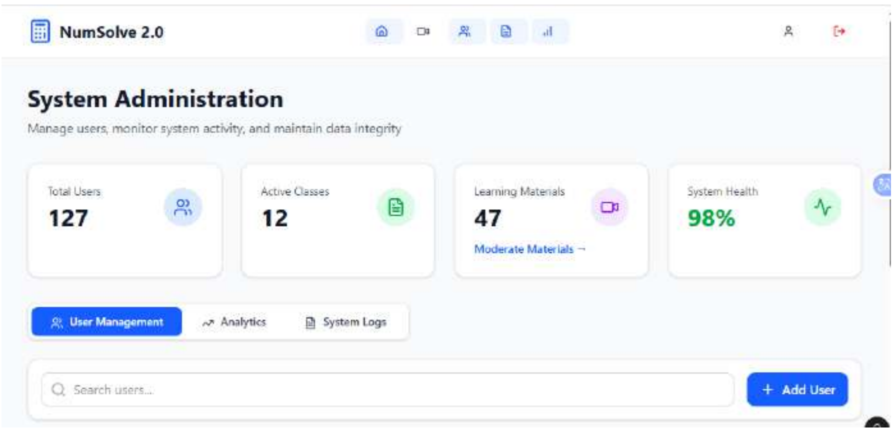


Figure 5.52: Educator Materials Management

System Analytics Dashboard

This dashboard visualizes core system metrics such as user sign-up trends and login activity. Admins can filter data by date range and user roles to have an idea about platform growth and usage patterns.

Figure 5.53: System Analytics Dashboard

Computation Activity Analytics

This dashboard focuses on a very particular aspect: computational load. It provides an overview of the frequency of numerical jobs, which allows the administrator to understand what moments of time are burdensome for the operating system.

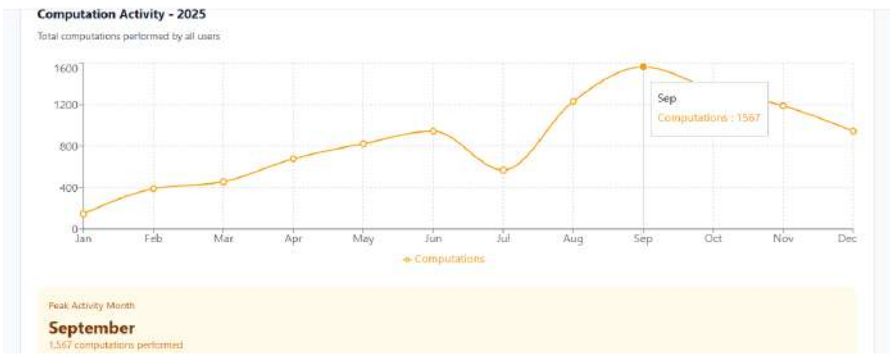


Figure 5.54: Computation Activity Dashboard

Gamification Analytics

Special analytics view tracking engagement with the new gamification features; shows trends of creating quizzes, participation rates by category, and overall engagement KPIs.

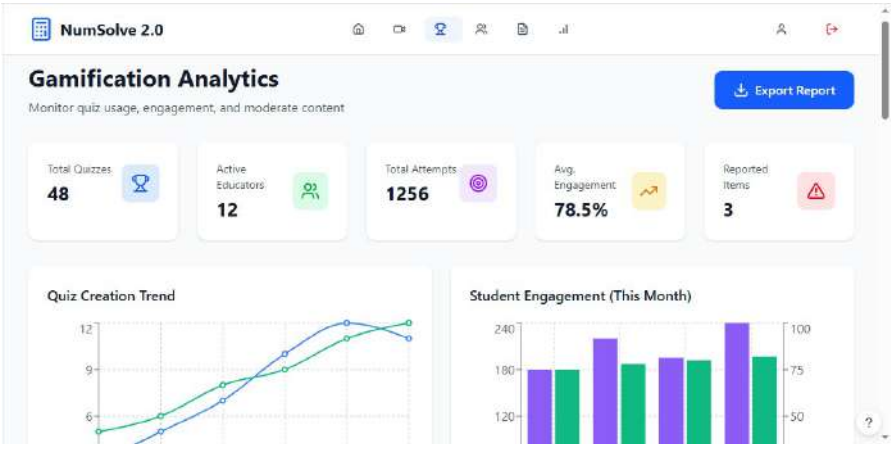


Figure 5.55: Gamification Analytics

System Administration Learning Library

This perspective helps in the administration of the content, and the administrators can review the learning materials uploaded and remove any inappropriate content, having the authority to do so.

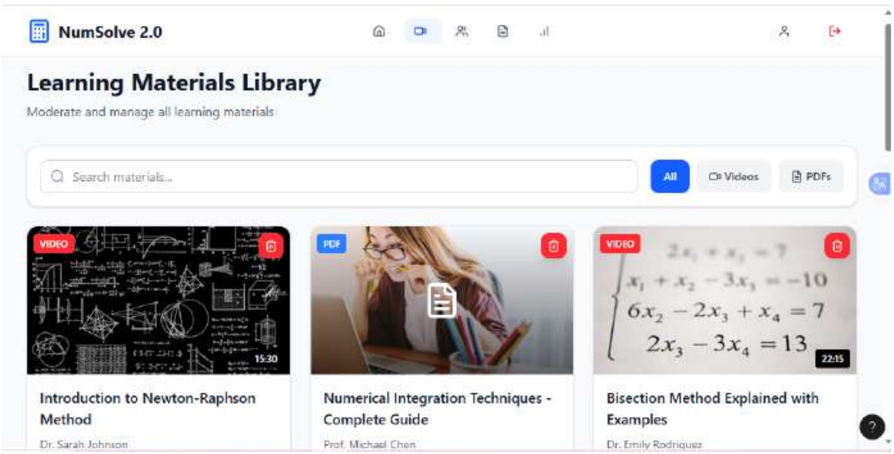


Figure 5.56: Learning Library Administration

5.8.3 Screen Objects and Actions

Table 5.15: Screen Objects and Actions - Authentication & User Management

Page Name	**Screen Object**	**User Action**	**System Response**
Landing Page	Sign In button	Click	Redirects to Login screen
Login Screen	Email field	Text input	Accepts email for authentication
	Password field	Text input	Accepts password for login
	Remember Me checkbox	Check/Uncheck	Stores session data locally
	Sign In button	Click	Validates credentials and opens dashboard
Registration Screen	Role selector	Select option	Sets user role (Student/Educator)
	Full Name field	Text input	Captures user's full name
	Email field	Text input	Captures email address
	Password fields	Text input	Sets and verifies password
	Create Account button	Click	Creates account and redirects to login
Student Dashboard	New Computation button	Click	Opens numerical computation interface
	Get Recommendation button	Click	Opens method recommendation input
	Navigation menu	Click	Navigates to different sections
	Statistics widgets	View	Displays user activity metrics
User Profile	Edit Profile button	Click	Enables profile editing mode
	Profile fields	Text input	Allows user to update profile details
	Save Changes button	Click	Updates user profile in database

Table 5.16: Screen Objects and Actions - Computation & Analysis

Page Name	Screen Object	User Action	System Response
Numerical Interface	Method category dropdown	Select option	Filters methods by category
	Specific method selector	Select option	Selects numerical algorithm
	Equation input field	Text input	Accepts mathematical equation
	Parameter fields	Text input	Accepts computation parameters
	Compute button	Click	Executes method and displays results and graph
	Save button	Click	Stores computation in database
Graph Visualization	Graph area	Hover	Displays coordinate values
	Zoom controls	Click	Zooms in or out for detailed view
	Legend items	Click	Toggles visibility of graph elements
Method Recommendation Input	Equation/Function field	Text input	Accepts problem for analysis
	Problem Type dropdown	Select option	Specifies problem category
	Desired Accuracy selector	Select option	Sets tolerance level
	Get Recommendation button	Click	Triggers rule-based recommendation
Method Recommendation Result	Use This Method button	Click	Applies recommended method
	Alternative method cards	Click	Selects an alternative method
	Advantages list	View	Displays advantages of the method

Table 5.17: Screen Objects and Actions - Class Management

Page Name	Screen Object	User Action	System Response
Classes (Student)	Search classes bar	Text input	Filters available classes list
	Join Class button	Click	Opens dialog to enter class code
	Class card	Click	Navigates to selected class details
	Instructor name link	Click	Opens instructor profile
Classes (Educator)	Create Class button	Click	Opens new class creation form
	Search classes bar	Text input	Filters educator's classes
	Class table row	Hover/View	Displays class details
	Edit icon	Click	Opens class settings
	Delete icon	Click	Deletes class after confirmation
Educator Dashboard	New Assignment button	Click	Opens assignment creation interface
	Upload Material button	Click	Opens material upload interface
	Activity statistics	View	Displays educator activity metrics

Table 5.18: Screen Objects and Actions - Records, Reports & Learning

Page Name	Screen Object	User Action	System Response
Computation Records	Search bar	Text input	Filters records by criteria
	Table rows	Click	Selects computation record
	View icon	Click	Displays detailed record
	Edit icon	Click	Opens record in edit mode
	Delete icon	Click	Deletes record from database
Report Generation	Computation dropdown	Select option	Selects computation for report
	Include options checkboxes	Check/Uncheck	Chooses report components
	Generate Preview button	Click	Generates report preview
Report Preview	Report area	View	Displays formatted report
	Export as PDF button	Click	Downloads PDF report
	Export as CSV button	Click	Downloads CSV report
Learning Library	Material cards	Click	Opens learning material viewer
	Filter dropdown	Select option	Filters materials by category
	Search field	Text input	Searches materials by keyword

Table 5.19: Screen Objects and Actions - Gamification & Quizzes

Page Name	Screen Object	User Action	System Response
Gamified Quizzes	Statistics tiles	View	Displays points and ranking
	Search quizzes bar	Text input	Filters quizzes by name
	Category/Difficulty filters	Select option	Filters quizzes by criteria
	Quiz card	Click	Starts quiz attempt
Leaderboard	Top 3 cards	View	Displays top-ranked users
	Rank list	Scroll	Displays full ranking list
	User profile card	Click	Opens user public profile
Quiz Management (Educator)	Create Quiz button	Click	Opens quiz creation interface
	Statistics tiles	View	Displays quiz statistics
	Visibility dropdown	Select option	Filters quizzes by visibility
	Edit Quiz icon	Click	Opens quiz editor
	Analytics icon	Click	Displays quiz analytics

Table 5.20: Screen Objects and Actions - Administration & Analytics

Page Name	Screen Object	User Action	System Response
System Administration	User Management button	Click	Opens user management panel
	System Logs button	Click	Displays system logs
	Analytics button	Click	Opens analytics dashboard
	System health indicators	View	Displays system performance metrics
System Analytics	Date range selector	Select option	Updates analytics charts
	User role filter	Select option	Filters analytics by user role
	Chart interaction	Hover	Displays detailed statistics
Gamification Analytics	Export Report button	Click	Downloads analytics report
	KPI tiles	View	Displays engagement metrics
	Quiz creation trend chart	Hover	Displays trend data
	Student engagement chart	Hover	Displays engagement levels
	Reported items widget	Click	Opens moderation queue
Computation Activity	Month selector	Select option	Updates activity statistics
	Peak activity indicators	View	Highlights peak usage periods
Library Administration	Review button	Click	Opens material review interface
	Delete button	Click	Removes inappropriate content

5.9 Discussion

As described in the previous chapter, the design phase of NumSolve 2.0 is a significant step in the progress of the project. It establishes the foundation for the future development of the technical architecture used to execute the theoretical requirements. The decision to create the application using a three-tiered architecture is pivotal to its overall success. The architecture separates three main components of an application: user interface (UI), application logic (AL), and data management (DM). This architecture allows for independent versions of each component to function separately from one another. Therefore, changes made to front-end components will not impact the performance of the back-end (core mathematical engine) which handles rule based method recommendations, iterates large amounts of numeric data in multiple iterations and provides applicable recommendations based on those iterations. A result of using a three-tiered architecture is that it provides the scalability that is crucial to supporting future enhancements needed for the ongoing number-based algorithmic innovations that will create a sustainable platform for all users maintaining the quality of the user experience will accomplish this objective.

This also reflects a user-centric design that seeks to make the connection between mathematical theory and real-world use explicit. The real-time graph visualization, gamified quiz elements, and intelligent recommendation design module allow for a much more complete form of learning experience than simply functioning as a mathematical calculator would provide. By providing separate dashboards for Students, Educators and Administrators, the design builders exhibit an exceptional amount of consideration for the overall architecture of the platform's ecosystem. Additionally, having the underlying database designed to the Third Normal Form (3NF) allows for a high degree of integrity to be maintained while enabling large data sets and user records to be processed with virtually no redundancy. All of the design choices illustrated herein provide a stable, efficient, and pedagogically sound platform that achieves the primary goals associated with the NumSolve 2.0 Project.

5.10 Summary

The technical blueprint for the NumSolve 2.0 system has been thoroughly documented in Chapter Five. This chapter provides a thorough technical description of the architectural, logical, and visual aspects of the software. A brief overview of the design phase and a high-level description of system architecture was presented at the beginning of the chapter, and this was followed with justification for the use of a three-tier model from an architectural perspective based on its modularity and maintainability. After establishing the architectural foundation for the development of the software, the chapter went on to outline the architectural structure of the software with package diagrams and an extensive database design, using entity relationship modeling and normalization techniques to ensure that the integrity of the data collected will be maintained.

A large portion of the chapter was devoted to the UI/UX design of the User Interface (UI) and User Experience (UX). The use of detailed screen storyboards and an extensive mapping of screen objects to system actions has allowed the document to visualize how different types of users will interact with the core features of the platform, including numerical computation, gamification, and administrative analytics. By completing all technical specifications for the project, there is now a complete technical framework to ensure that all functional and non-functional requirements for the project will be met. With the design blueprints completed, this research will proceed to the next phase of the project as outlined in Chapter Six: System Implementation, where these logical models will be converted into a high-performance executing software system.

CHAPTER 6

SYSTEM IMPLEMENTATION

The project implementation stage of NumSolve 2.0 is the focus of this chapter. It explains how the architecture and database design developed in the previous stage were converted into a working software application. It details the development environment used as well as which programming languages were used to create the core modules of the system. The chapter also documents how the coding was done and how hierarchical structure of the application was developed so that it serves as a complete record of how the technical specifications of the project were turned into an intelligent cross-platform software application.

6.1 Introduction

At the end of the software development life cycle (SDLC), we have completed our theoretical designs and are now ready to implement them into a live or operational product. For NumSolve 2.0, this process involves coding all three tiers of the architecture systematically, creating complex numerical mathematical algorithms, and configuring the MySQL database to store data for future reference. The primary objective for the final software product is to ensure it meets all functional specifications as defined during the requirements phase, along with meeting excellent coding standards such as efficient, secure, and cross-platform compatible code.

In this stage of the development process, the team is focused on implementing the business logic for the various engines (mathematical, Rule-Based Decision

Support Module) and calculating algorithms (Bisection, Newton-Raphson, ODEs). This implementation phase will also include the front-end interface using JSP and JavaScript to provide an interactive and responsive solution. This section describes the tools and technologies that have been used, as well as the methodologies employed during the implementation phase, in order to ensure a smooth transition from a static design storyboard to a dynamic and intelligent system. By documenting the implementation, the project aims to achieve a certain level of technical transparency, which will help with further maintenance, further development, or academic research.

6.2 System Hierarchical Menus

The user interface of the **NumSolve 2.0** system is designed with a role-based navigation architecture to cater to the specific workflows of Students, Educators, and Administrators. The following hierarchical structures outline the modules available to each user type, ensuring an intuitive user experience while maintaining strict access control boundaries.

6.2.1 Student Dashboard Hierarchy

The Student Dashboard is tailored to facilitate learning, continuous assessment, and mathematical problem-solving. Its primary focus is to provide quick access to educational resources and the core computation engine.

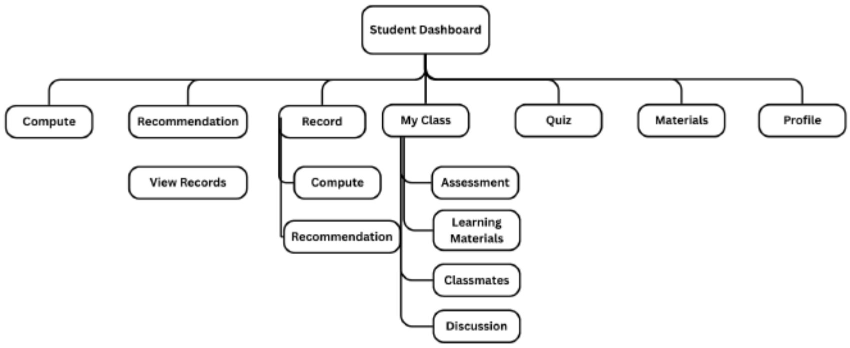


Figure 6.1: Hierarchical Menu Structure for Student Role

- **Compute:** Direct access to the system's numerical method engine, allowing students to input equations, set parameters, and generate mathematical solutions and graphs.
- **Recommendation:** A module providing personalized suggestions to the student based on their system usage.
  - **View Records:** Allows students to browse the history of recommendations provided to them.

- **Record:** A comprehensive historical log of the user's past activities within the platform.
  - **Compute:** A sub-module displaying the history of past mathematical calculations, including inputs, iterations, and final results.
  - **Recommendation:** A historical view of past system recommendations.
- **My Class:** The central hub for course-specific activities, showing the classes the student is enrolled in.
  - **Assessment:** Access to quizzes, assignments, and evaluations specific to a selected class.
  - **Learning Materials:** A repository of notes, PDFs, and resources uploaded by the class educator.
  - **Classmates:** A directory of peers enrolled in the same class to facilitate networking.
  - **Discussion:** The communication portal for group messages and direct peer-to-peer or student-to-educator interactions.
- **Quiz:** A general directory for public or platform-wide assessments not strictly tied to a specific enrolled class.
- **Materials:** A general library of open-access educational materials available to all students.
- **Profile:** Account management settings where students can update their personal information, biography, and profile picture.

6.2.2 Educator Dashboard Hierarchy

The Educator Dashboard builds upon the student features but introduces extensive content creation and classroom management capabilities. It empowers educators to distribute knowledge and evaluate student performance efficiently.

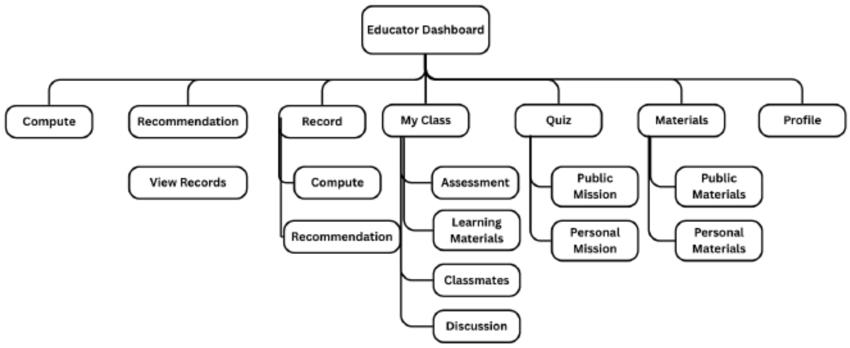


Figure 6.2: Hierarchical Menu Structure for Educator Role

- **Compute:** Access to the core numerical computation engine to test formulas or demonstrate calculations.
- **Recommendation:** System-generated insights or suggestions relevant to the educator.
  - **View Records:** A log of past recommendations viewed by the educator.
- **Record:** A historical log of the educator's computational activities.
  - **Compute:** Saved logs of past mathematical operations performed by the educator.
  - **Recommendation:** History of system suggestions.

- **My Class:** The classroom management interface where educators can oversee the virtual rooms they have created.
  - **Assessment:** Tools to create, distribute, and grade class-specific quizzes.
  - **Learning Materials:** An interface to upload and manage class-specific educational files and documents.
  - **Classmates:** A roster of all students currently enrolled in the educator's class.
  - **Discussion:** A moderation and communication hub for class-wide broadcasts and direct student inquiries.
- **Quiz:** The assessment management hub, categorized by visibility.
  - **Public Mission:** Quizzes created by the educator that are published globally for any platform user to attempt.
  - **Personal Mission:** Private quizzes created specifically for targeted classes or distinct groups.
- **Materials:** The resource management hub for the educator's uploaded files.
  - **Public Materials:** Educational resources shared openly with the entire NumSolve 2.0 community.
  - **Personal Materials:** Private files reserved for specific classes or personal reference.
- **Profile:** Account settings for updating the educator's academic credentials, department, and contact details.

6.2.3 Administrator Dashboard Hierarchy

The Administrator Dashboard bypasses the educational workflows to focus strictly on platform oversight, system health, and data moderation. It provides a macro-level view of NumSolve 2.0.

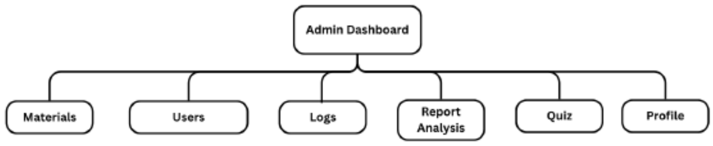


Figure 6.3: Hierarchical Menu Structure for Administrator Role

- **Materials:** A global moderation view of all learning materials uploaded across the system, allowing the admin to review, approve, or delete resources to maintain quality standards.
- **Users:** The central user management module to oversee all registered accounts (Students and Educators), adjust role permissions, and handle account suspensions or password resets.
- **Logs:** A technical audit trail displaying system events, user login histories, and computation execution records to aid in troubleshooting and security monitoring.
- **Report Analysis:** A data visualization module providing statistical insights into platform usage, quiz completion rates, and overall user engagement metrics.

- **Quiz:** A global moderation view of all assessments created on the platform, ensuring content remains appropriate and technically functional.
- **Profile:** Administrative account settings and security configurations.

6.3 System Development

The development of the NumSolve 2.0 application was based on system architectural and design specifications outlined in previous chapters of this book. During the implementation phase, the proposed design was translated into a fully functional web application that provides the capability to perform numerical analysis calculations; conduct educational activities; and make intelligent method recommendations. The application was developed using Java Server Pages (JSP), Java Servlets, and the MySQL Database Management System; these technologies were chosen due to their reliability, scalability, and compatibility for developing web applications.

The implementation process utilized a Three-Tier Architecture, which has a Presentation Layer, an Application Layer, and a Database Layer. The Three-Tier Architecture promotes modular design by separating the User Interface components from the Application Logic components and the Data Management components. Therefore, it makes the application easier to maintain, enhance, and troubleshoot.

6.3.1 Presentation Layer Implemention

The Presentation Layer interacts with the users of the system. The technology used in the creation of the layer includes JSP, HTML, CSS, Bootstrap and JavaScript which allow the creation of responsive web pages which are easy for users to navigate. The Presentation Layer features a number of interfaces such as; login page, user registration page, dashboard, numerical computation forms, pages for learning materials, quiz interface and administration management system.

Usability and accessibility were important considerations for the design of the interface for all users of the system. The interface was designed with a hierarchical menu structure to allow organized access to the system functionalist based on the users' role, i.e., there are different menus and permissions available to students, educators and system analysts based on their individual role within the system.

6.3.2 Application Layer Implementation

The Application Layer of NumSolve 2.0 contains the core functionality. Implementation was carried out by using Java Servlets and the business logic (processing user requests and executing computational tasks).

The Numerical Computation Module is one of the primary components implemented within the Application Layer. This module supports performing different types of numerical methods such as root-finding, interpolation, numerical differentiation, numerical integration, and ordinary differential equations. All user inputs are validated before processing occurs and results are displayed with computational information.

The Rule-Based Decision Support Module is another important component of the Application Layer. This component was developed and implemented by creating predefined IF-THEN rules that evaluate the input characteristics of a numerical problem to recommend a particular numerical method. The recommendation mechanism can help users (especially beginners) to choose which technique to use, without having to possess extensive prior knowledge of the numerical analysis.

The Application Layer also includes the Learning and Assessment Module. This component has been developed to manage learning materials, tutorial guides, quizzes, and educational resources for students to have access to supporting materials while engaged in performing numerical computations on the same platform.

Additionally, as part of this project, another set of visual tools has been implemented in order to help users understand the relationship of a sequence of numbers visually. The Visualization Module provides a visual representation of functions, iterative convergence behavior, and results of computation. The ability to see the results of computations visually has assisted users in obtaining and determining the correctness of the numerical solution as well as in viewing how well the algorithms perform based on the visual representation provided by the Visualization tools.

6.3.3 Database Layer Implementation

The implementation of the Database Layer was done with MySQL. MySQL is a durable storage database and allows for durable storage of information on a networked device. The implementation of the Database Layer required the creation of several database tables for storing information related to user account management, computation history, and computation learning material, quiz information, class management, and general system performance data.

Data retrieval and manipulation between the application and the database occurs using the Java Database Connectivity (JDBC) API. CRUD operations have been implemented to provide the ability for efficient creation, retrieval, updating, and deletion of data from the various components of the overall system. To minimize the amount of redundancy in the database, proper database normalization techniques were utilized. This will also improve the performance of the database as a whole.

6.3.4 Security Implementation

The security features of the system were designed to protect the identity of users and provide a mechanism for ensuring the secure operation of the system.

The use of different methods of authentication have been incorporated so that each user can be verified before they gain access to a protected resource. All user accounts have been given strong passwords and the use of proper session management techniques has ensured that no user has been able to gain unauthorized access to the protected resource after they logged in.

6.4 Discussion

The NumSolve 2.0 has turned this proposal into a fully functioning web-based tool that can be used for both numerical analysis and educational purposes. All of the project's features related to numerical computation, method selection, visualization, and education now exist in one environment, eliminating the need for students to use multiple applications.

An important feature of the implementation phase is the Rule-Based Decision Support Module, which assists users in selecting appropriate numerical methods according to a set of established guidelines. This feature makes using the system easier for new users and also fulfills part of the project's education purpose.

The application of a Three-Tier Architecture was demonstrated to be an effective way to separate presentation, application logic, and database functionality within the system. This methodology provided better organization, maintenance, and future expandability of the entire system.

There were challenges with the development of the system, specifically related to the integration of different modules and maintaining consistent communication between the user interface, computation engine, and database. However, the completion of the project establishes that the technologies and architecture chosen will create an accessible and intelligent numerical analysis system.

6.5 Summary

The development of NumSolve 2.0 was undertaken by designing it according to the design specifications from Chapter 2. NumSolve 2.0 was built upon Java, JSP and MySQL and other related web technologies utilising a Three-Tier Architecture.

The components included in the development of NumSolve 2.0 were the following: Presentation Layer, Application Layer, Database Layer, Numerical Computation, Rule-based Decision Support, Visualisation, Learning Resources, Quizzes, and User Management. To provide a secure environment for users, security was implemented through user authentication, input validation and role-based access control.

The implementation of NumSolve 2.0 created a web-based application that is both functional and maintainable and that is fulfilling the overall objectives of the project. The following chapters will conclude this study with an overview of the contributions to the field, limitations of this study and plans for future developments.

CHAPTER 7

CONCLUSION

This chapter summarises the entire NumSolve 2.0 Project and describes the process of developing NumSolve 2.0 from its inception (the initial elicitation of requirements) through to its final state (the implementation of the completed system). It discusses how this research has made important contributions to the world of education and (numerical) analysis, identifies various technical and operational limitations that were faced during the course of this project, and provides strategic recommendations for improving the system in the future, in order to continue to be valid and evolving tools for students and researchers.

7.1 Introduction

In conclusion, this chapter presents a synthesis of the entire research and development of NumSolve 2.0: a cross-platform intelligent numerical analysis suite. A key focus of this project has been to link theoretical mathematical concepts with applying them computationally, through the introduction of an accessible web-based application that provides intelligent guidance.

The core goal of the project was to create a platform that has both computational capability and provides support for pedagogical purposes. The example of a successful integration between a three-tier architecture and a rule-based decision support system has fulfilled the project objective of creating this platform.

The chapter summarises the results of the development process, determines the level to which the original project objectives were achieved and describes how the system can help support learning in the university environment (examples include gamifying student engagement and giving teachers better management tools). Additionally, this chapter outlines limitations present within the current state of the project, provides the basis for future development areas (such as using artificial intelligence to develop enhanced error analysis capabilities), and serves as a guide for future development possibilities for NumSolve to become a long-term digital resource in mathematics and computer science.

7.2 Project Contributions

The development of NumSolve 2.0 supports numerical analysis education by creating a unified web-based platform that combines learning materials, numerical computation, and intelligent recommendations of numerical techniques. Different from traditional numerical software that performs only calculations, NumSolve 2.0 allows users to solve problems while learning as they go. In turn, this will help users understand the concepts and applications surrounding numerical methods.

An important part of the project is the installation of the Rule-Based Decision Support Module. This module will enable users to choose the most suitable numerical techniques to perform calculations for a given problem based on its particular features. When users have access to recommendations from the system, users who have little experience can access more of the information on how to successfully apply numerical techniques to solve a given numerical problem.

Additionally, the incorporation of visualization tools, quizzes, and learning materials into the platform will increase the educational usefulness of NumSolve 2.0 by promoting self-directed learning and improving users' grasp of concepts through practice, and encouraging interactive engagement with the learning process. Instructors will also be able to utilize the system to effectively supervise students' progress, and thereby better facilitate classroom activities.

From a software engineering perspective, the system is an excellent example of the successful application of a Three-Tier Architecture utilizing Java, JSP and MySQL technologies. The modular design of the application will provide for easy maintainability, interoperability, and flexibility which will create a strong foundation for future enhancements and growth of the overall system.

7.3 Project Constraints

Although NumSolve 2.0 was successfully implemented, there were numerous limitations discovered throughout the project. One limitation is that the Rule-Based Decision Support Module is based on predefined rules. The recommendation mechanism works well for typical numerical problems; however, it is unable to learn from user actions or modify itself to accommodate new situations.

Another limitation is that there is a dependency on an internet connection to use NumSolve. It is built as a web application, and users would need internet access to use NumSolve's services. This could cause access issues if users are in areas with limited or no internet service.

Currently, the system only supports certain numerical methods that fit within the project's scope. The project does not currently offer any advanced numerical methods such as optimisation algorithms, matrix decomposition techniques, or finite element analysis.

Additionally, the version being used today is primarily focused on achieving project goals. Other advanced features such as artificial intelligence-based recommendations; mobile application support; and cloud-based collaboration features are not part of the current project and will be developed at a later time.

7.4 Future Work

The functionality and effectiveness of NumSolve 2.0 could be improved in a number of ways. One way that could be accomplished would be via the incorporation of artificial intelligence (AI) and machine learning (ML) methodologies to provide recommendations to the users based on their behaviours, patterns of learning, and previous computational activities.

Another recommendation to improve NumSolve 2.0 would be to expand the number of supported numerical methods to include more complex numerical techniques for solving engineering and general numerical problems, such as optimisation algorithms, matrix decompositions, finite difference methods (FDM), and finite element analysis (FEA).

Mobile applications for Android and iOS could also be developed for future releases in order to increase the accessibility of the software and make it easier for users to compute values, view learning resources, etc., on their mobile devices.

Collaboration features such as cloud-based collaboration may also be incorporated in order for students and instructors to interact with each other in real-time thus, enabling collaborative learning and discussion of student projects and other classroom activities within the platform.

Finally, sophisticated learning analytics and performance monitoring tools would provide educators with better insights into the progress of their students and would help educators identify and support students who require assistance in order to succeed.

7.5 Summary

This chapter provided an overview of the conclusions drawn from the entire NumSolve 2.0 project, including the project's contributions, constraints, and possible avenues for future development. The project was successful in meeting its objectives by providing an integrated web-based platform for numerical analysis that includes the development of a numerical computing environment with integrated learning materials, visualization, and a Rule-Based Decision Support Module in one place.

In addition to allowing users to benefit from an easier to access and more interactive numerical based education, the project also provides teachers with the ability to effectively manage their teaching and learning activities. Even though the project may have limitations, such as being limited to predefined rules or a small number of numerical methods, the finished product validates the possibility of using a singular application to combine both computational capabilities and educational functions.

Future enhancements, such as incorporating artificial intelligence, developing a mobile application, enabling cloud based collaborative working, and by adding other numerical methods to the application may allow for continued improvements. Overall, the NumSolve 2.0 project provides a strong starting point for continued development and will provide educators with valuable assistance when teaching and learning about numerical analysis.

REFERENCES

APPENDICES

BIODATA OF THE AUTHOR

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As the developer of NumSolve 2.0, Ameer has focused on bridging the gap between high-level numerical analysis and intuitive educational tools. His work emphasizes the transition from expensive, platform-dependent software to open-access, web-based solutions. Beyond his technical pursuits, Ameer is an active member of the Kelab Pengacaraan dan Komunikasi (GEMERSIK), where he has honed his skills in communication and presentation—traits that are reflected in his commitment to making complex mathematical concepts more "communicable" through interactive visualization.

His research interests include web-based application architecture, intelligent decision support systems, and the democratization of educational technology. He can be reached for inquiries or collaboration at muqrieameer@gmail.com.

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